Mach-26
Critical Design Review

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Category 1

Contents

1. Introduction

1.1 Mission Statement

To design, build, and successfully launch a model rocket capable of reaching a target apogee of 450 meters, carrying a camera-equipped CanSat to capture and store continuous in-flight video, while safely transporting and recovering a fragile extra payload (quail egg) besides the CanSat. This mission aims to demonstrate reliable payload and rocket integration, data acquisition, payload protection, and recovery under real flight conditions, in full compliance with the UKRA Model Rocket Safety Code.

Since the Preliminary Design Review (PDR), the mission statement has been refined to improve clarity and a better reflect of the technical objectives of the project.

1.2 Objectives

ID	Title	Description
OBJ-1	Target Apogee Accuracy	The model rocket system shall reach an apogee as close as possible to 450 meters, keeping a stable flight in accordance with the UKRA Model Rocket Safety Code.
OBJ-2	CanSat Video Acquisition During Flight	The CanSat shall be capable of recording and storing continuous video throughout the entire flight, from liftoff to recovery.
OBJ-3	CanSat Telemetry, Tracking, and Data Storage	The CanSat shall integrate altitude and GPS sensors, also acquire, store, and provide access to flight data, including a redundant tracking capability.
OBJ-4	Safe Transport and Recovery of Additional Payload	The payload system shall ensure the safe transport and recovery of a quail egg, keeping its structural integrity during ascent, descent, and landing, without compromising flight safety.
OBJ-5	Controlled Descent and Recovery System	The recovery system shall limit the descent rate to safe values and ensure that both the rocket and the CanSat land within drift radius of 400 meters.
OBJ-6	Compliance, Safety, and Verification	All rocket and CanSat subsystems shall be designed, integrated, and verified in accordance with the UKRA Model Rocket Safety Code, including structural integrity, stability margins, electrical safety, and recovery system reliability.

2. Launch Vehicle

2.1 Launch Vehicle Overview

The launch vehicle is a model rocket designed for CANSAT deployment, integrating a 3D-printed aerodynamic nose cone, internal payload support structure, and a parachute-based recovery system for safe descent.

2.1.1 Modifications

Detailed Modifications and Component Evolution

Following the Preliminary Design Review (PDR), the M.I.L.A.N.E.S.O. launch vehicle underwent a comprehensive structural and mass-property optimization phase. This iterative process was driven by the requirement to increase the safety factor of the aerodynamic surfaces and to refine the stability margin. The final configuration achieved a mass reduction of 51 g, resulting in a launch-ready mass of **1091 g.**

A. Redesigned Components

- **Primary Aerodynamic Surfaces (Fins):** The initial proposal of solid cedar wood was rejected due to its inherent orthotropic limitations and low shear modulus across the grain. The fins have been re-engineered as **Custom Quasi-Isotropic Cedar Laminates**.

Technical Specification: Each fin is composed of multiple cedar veneers, with the grain orientation of adjacent layers offset by 90 degrees. This creates a balanced laminate that exhibits uniform mechanical properties in the longitudinal and transverse axes.

Manufacturing Process: The veneers are bonded using a high-viscosity, structural-grade epoxy resin. The assembly is subjected to a **uniform compression cycle** using a precision mechanical press and industrial G-clamps. This controlled clamping pressure ensures a minimal bond-line thickness and maximum fibre-to-matrix ratio, effectively eliminating the risk of inter-laminar delamination or aeroelastic flutter at the projected maximum velocity of **102 m/s**.

- **Ogive Recovery System Interface (Nosecone):** The nosecone architecture was transitioned from an ambiguous "solid infill" model to a **High-Density Shell Membrane** design.

Technical Specification: Manufactured via Fused Deposition Modelling (FDM) in Polyethylene Terephthalate Glycol (PETG), the component features a **5.0 mm perimeter wall thickness**.

Engineering Justification: This 5.0 mm "solid shell" provides the necessary hoop strength to resist stagnation pressure and skin friction heating at Max-Q. By maintaining a hollow core, the component mass was fixed at **180 g**, allowing for precise ballast control. This ensures the Centre of Gravity (CG) remains at 38.1 cm, providing a stable margin of **1.84 calibres**.

B. Retained Components

The following subsystems were validated during the PDR and remain unchanged in the final flight configuration:

- **Airframe Structure:** The 3-inch diameter spiral-wound Kraft glassine body tube was retained due to its excellent strength-to-weight ratio and proven resistance to buckling under the 102 N peak thrust.

- **Propulsion Module:** The Aerotech G80T-7 (Single Use) remains the primary motor, providing the necessary impulse to achieve a safe off-rail velocity of **13.1 m/s**.
- **Recovery Hardware:** The Topflight Recovery PAR-18 (18-inch) ripstop nylon parachute and the 12-strand Kevlar shock cord remain the baseline for the recovery sequence.

C. Strategic Impact on Budget and Timeline

- **Financial Optimisation:** The transition to in-house cedar lamination represents a strategic cost-saving measure. By utilising raw timber veneers and structural epoxy already available in the laboratory, the team avoided the significant lead times and high procurement costs of imported aerospace-grade G10 fibreglass or birch plywood.
- **Work Breakdown Structure (WBS) Impact:** The custom lamination process introduced a specific **24-hour isothermal curing window** into the manufacturing timeline.

2.2 Launch Vehicle Design and Verification

2.2.1 Complete Design with Justifications

The M.I.L.A.N.E.S.O. launch vehicle is engineered as a single-stage, unguided suborbital rocket optimised for the Mach-26 450 m apogee target.

The following details the final component architecture:

- **Nose Cone:** An ogive-profile aerodynamic fairing manufactured via FDM 3D printing in PETG. It features a rigidly controlled 5.0 mm perimeter wall thickness and the internal volume is predominantly hollow to reduce weight, while preserving structural integrity through the controlled wall thickness. It integrates a 30 mm shoulder (64 mm outer diameter) for a secure friction fit with the primary airframe. Additionally, it includes an internal metallic eye bolt anchored at the base, serving as the attachment point for the shock cord.

Fig. 1. Cross-sectional view of the cone

- **Airframe Tubing:** The main fuselage is constructed from 3-inch (76.2 mm OD) spiral-wound Kraft glassine tubing. This provides a highly favourable strength-to-weight ratio and maintains RF transparency for the internal telemetry module.

- **Coupler Tubing:** Also manufactured from Kraft glassine, functioning as the structural interface between airframe segments and housing the payload. It features a 7.44 cm OD to match the main airframe's internal diameter seamlessly.
- **Fins:** A four-fin configuration manufactured from custom quasi-isotropic cedar laminate (cross-grain veneered and epoxy-cured).
- **Motor Tube:** A 29 mm internal diameter Kraft phenolic tube designed specifically to house the Aerotech G80T-7 Single Use motor, isolating the thermal and mechanical loads from the outer airframe.
- **Centering Rings:** Precision-cut structural rings (wood composite) that concentrically align the 29 mm motor tube within the 76.2 mm main airframe. They transfer the 102 N peak thrust directly into the body tube walls. Also, the forward centering ring acts as the primary anchor point for the 14 mm. In addition, the front centering ring acts as the main anchor point for the 14mm Kevlar elastic cord, thus performing the function of a bulkhead.
- **Motor Retainer:** A mechanical retention system affixed to the aft centering ring. As the G80T is a single-use motor without a flanged casing, this retainer positively locks the motor in place, preventing forward penetration during thrust and rearward ejection during the black powder delay charge deployment.
- **Launch Lugs / Buttons:** Two standard linear rail buttons rigidly mounted on the airframe, the first at the height of the center of gravity (38.4 cm from the nose tip) while the second is 21cm behind the first, this to interact with the nominal 2.0m launch rail, ensuring a zero-degree angle of attack until off-rail speed is reached.

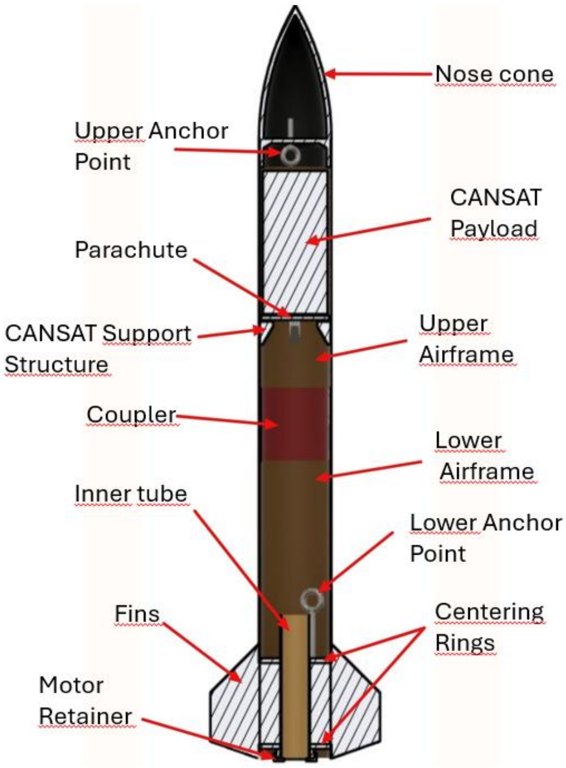


Fig. 2. Cross-section view of the rocket

2.2.2 Integrity of Design

Fin Shape and Size The fin geometry utilises an elliptical trapezoidal planform. This specific shape was selected to minimise aerodynamic drag while shifting the Centre of

Pressure (CP) sufficiently aft (to 52.1 cm from the nose tip). The sizing was iteratively optimised in the 6-DOF simulation to balance the aerodynamic restoring forces against the vehicle's mass distribution, culminating in a highly stable, final static margin of **1.84 calibres**. This ensures rapid dampening of any pitch or yaw moments post-rail exit without making the vehicle over-stable and susceptible to crosswind weathercocking.

Materials for Fins, Bulkheads, and Airframe Tubing Material selection strictly adhered to Mach-26 non-metallic safety regulations while maximising specific strength:

- **Fins:** The in-house cedar laminate effectively neutralises the orthotropic weakness of solid wood. By orienting the grain at 90-degree offsets and curing under mechanical pressure, the composite fins achieve the shear modulus required to resist aeroelastic flutter at the maximum velocity of 102 m/s.
- **Airframe Tubing:** Spiral-wound Kraft glassine was chosen over fibreglass to strictly maintain the mass budget. It offers exceptional longitudinal stiffness against the compressive loads generated during the 13.1 m/s rail exit and subsequent acceleration.
- **Center Ring:** Engineered wood composites are utilised to bear the immense tensile loads generated during parachute deployment. When the ejection charge fires, deploying the system at 12.1 m/s, the bulkheads must transfer the kinetic energy from the 14 mm Kevlar shock cord to the airframe without shearing.

Motor Mounting and Retention The integrity of the propulsion interface is paramount. The Aerotech G80T-7 generates a peak thrust of 102 N and an average thrust of 78 N over 1.71 seconds. This force is distributed through the motor tube into the aft and forward centering rings, which are bonded to the airframe using structural epoxy fillets. This prevents stress concentrations that could cause the airframe to buckle. The mechanical aft retainer ensures the motor casing remains locked inside the vehicle throughout the entire flight profile, neutralising the risk of ejecting hazardous debris over the range.

Mass Estimates A rigorous mass management protocol was enforced following the PDR. The total lift-off mass is precisely calculated at **1091 g**.

- **Propulsion:** The fully loaded G80T motor contributes 62.5 g.
- **Forward Structure:** The 5.0 mm PETG nosecone is calibrated to 170 g, providing critical forward ballast to maintain the 1.84 calibre static margin.
- **Payload:** The CanSat module accounts for a fixed 350 g.
- **Airframe & Recovery:** The remaining mass is distributed among the Kraft tubing, the custom laminate fins, the Top Flight PAR-18 parachute, and the 12-strand Kevlar tether. This optimised mass budget yields a high Thrust-to-Weight ratio of **7.29:1**, driving the rapid acceleration necessary to achieve the safe 13.1 m/s off-rail velocity.

2.3 Flight Safety Analysis

Extensive 6-DOF (Degrees of Freedom) flight simulations were conducted using OpenRocket to validate the aerodynamic stability and safety profile of the M.I.L.A.N.E.S.O. vehicle. The mass model was strictly calibrated to the final lift-off mass of **1091 g**.

2.3.1 Launch Day Conditions and Analysis

Given the geographical context of the Mach-26 competition (Scotland, UK), the launch vehicle must be capable of safe operation in variable maritime/coastal atmospheric conditions, specific to the geography of the launch site. The flight model has been analysed against the strict safety limitations outlined by the UK Rocketry Association (UKRA).

Wind represents the primary operational risk, as it introduces lateral loads and deviations in the trajectory. These loads are governed by dynamic pressure: $q = \frac{1}{2}\rho V^2$. For typical conditions in Campbeltown in June ($\rho \approx 1.23 \frac{kg}{m^3}$) and a maximum vehicle speed of 102 m/s: $q_{max} \approx 6500\ Pa$.

- **Wind Profile:** The vehicle is cleared for launch in sustained crosswinds of up to **20 mph (approx. 9 m/s)**.
- **Weathercocking Mitigation:** To prevent the vehicle from aggressively turning into the wind (weathercocking) and breaching range safety boundaries, a high Thrust-to-Weight (T:W) ratio of **7.29:1** was engineered. Driven by the Aerotech G80T-7 motor, the vehicle achieves a calculated **off-rail velocity of 13.1 m/s** (assuming a nominal 2.0-metre launch rail). This velocity ensures that the custom composite fins generate immediate and dominant aerodynamic restoring forces the moment the vehicle leaves the guide rail, $F_{aero} \propto V^2$, neutralising crosswind deflection.

2.3.2 Flight Analysis and Impact on Design

The predicted kinematic profile directly dictated several critical structural modifications to ensure the vehicle's survival and the public's safety:

- **Max-Q and Aerodynamic Heating:** The vehicle reaches a maximum velocity of **102 m/s (Mach 0.3)** shortly after motor burnout (1.71 s). *Impact on Design:* This high dynamic pressure invalidated the use of standard cardboard or solid wood fins. The airframe was thus upgraded to utilise quasi-isotropic cedar laminates (Adopting a design criterion $FS \geq 2$) and a 5.0 mm thick PETG nosecone to prevent structural deformation or aeroelastic flutter.
- **Thrust Loading:** The G80T motor produces a peak thrust spike of **102 N**. *Impact on Design:* This required the implementation of engineered wood centering rings and a mechanical aft-retainer to prevent the motor casing from penetrating the forward avionics bay or being ejected on the pad.
- **Apogee and High-Energy Deployment:** The predicted apogee is **476 m**, which represents a highly precise ballistic trajectory relative to the 450 m target. However, simulation indicates the optimum deployment delay is 8.17 seconds, while the selected motor has a fixed 7-second delay. Consequently, the ejection charge will fire prior to apogee, deploying the parachute at a velocity of **12.1 m/s**. *Impact on Design:* To safely dissipate the kinetic energy of this early deployment without causing a catastrophic structural failure (*zippering*), the recovery tether was upgraded to a high-tensile **12-strand Kevlar shock cord (14 mm width)**, securely anchored to reinforced internal bulkheads.

2.3.3 Launch Vehicle Stability

The M.I.L.A.N.E.S.O. demonstrates a highly robust stability profile, designed to self-correct aerodynamic disturbances instantaneously without becoming over-stable.

- **Static Margin:** At the moment of lift-off, the Centre of Gravity (CG) is located at 38.1 cm from the nose tip, and the Centre of Pressure (CP) is located at 52.1 cm. This geometry provides a safe static margin of **1.84 calibres**, which falls perfectly within the optimal 1.5 to 2.5 range required by Mach-26 guidelines.
- **Dynamic Progression:** The Aerotech G80T motor contains approximately 37.5 g of solid propellant. As this propellant is consumed during the 1.71-second burn phase, the aft mass decreases, causing the CG to shift forward. As a result, the static margin progressively increases during powered flight. This dynamic shift ensures the vehicle becomes increasingly stable exactly when it experiences the highest aerodynamic forces, safely maintaining a vertical trajectory towards apogee.

2.3.3 Drift Analysis and Recovery Zone

- **Drift:** The horizontal displacement of the launch vehicle during the descent phase represents one of the primary flight safety constraints, as the system must remain within the maximum recovery radius of 400 m established by the mission requirements.
  Is estimated as a function of wind velocity and descent time, using the selected parachute of 18" (46 cm diameter), being $V_t = 10.2\frac{m}{s}$ and the descent time of 44 s, with height of 450 m. Wind conditions representative of Campbeltown during summer were considered wind speed of 5 m/s under nominal conditions and 7 m/s under limiting conditions, resulting in a drift of **220 m** for the nominal case and **308 m** for the limit case. The vehicle is cleared for launch in sustained crosswinds of up to 9 m/s, but that would be the edge of the safety limit. The predicted drift remains within the allowable recovery radius of 400 m under both nominal and worst-case operational wind conditions. This indicates that the launch vehicle recovery system satisfies range safety constraints and ensures containment within the designated landing area. To ensure compliance with recovery limits, launch operations are restricted to wind speed $\leq$ **7 m/s.**
- **Design Implications:** A trade-off between descent velocity and drift was identified, increasing parachute area would reduce landing velocity but increase horizontal displacement. The selected parachute configuration prioritises containment within the recovery zone.

It is important to note that this analysis corresponds exclusively to the **launch vehicle recovery system**. The CanSat payload employs an independent descent system, which is analysed separately.

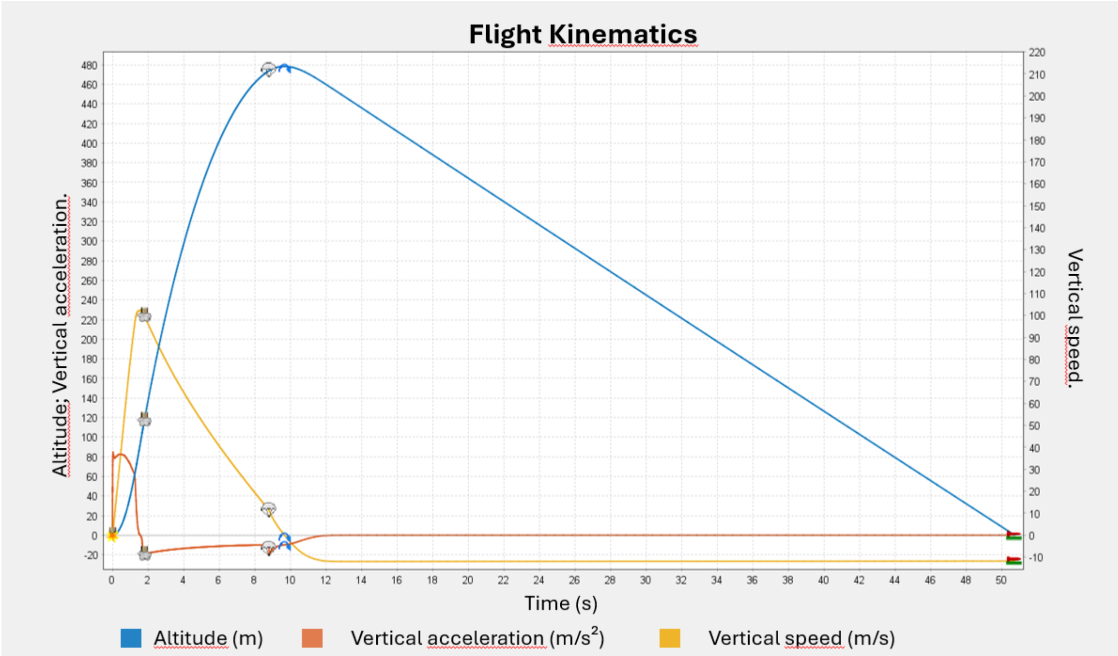


Fig 3. Flight Kinematics – Altitude, Acceleration and Vertical speed.

Fig. 3. Flight Kinematics – Altitude, Acceleration and Vertical speed

2.4 Motor Selection

2.4.1 Primary Motor Specification and Justification

The primary propulsion system for the M.I.L.A.N.E.S.O. launch vehicle is the **Aerotech G80T-7 (Single Use).**

- **Performance Metrics:** Peak Thrust: 102 N | Average Thrust: 78 N | Total Impulse: 133 Ns | Burn Time: 1.71 s.
- **Design Justification:** With the final lift-off mass of **1091 g**, this motor provides a high Thrust-to-Weight (T:W) ratio of **7.29:1**. This ensures a safe off-rail velocity of **13.1 m/s** (on a 2.0-metre rail), giving the fins immediate aerodynamic authority and preventing weathercocking in the UK's variable wind conditions.
- **Delay Selection:** The 7-second delay is the optimal commercial choice to ensure deployment occurs safely near the predicted **476 m** apogee.

2.4.2 Status of Motor Procurement

The team has successfully mitigated the risks associated with international propellant transport regulations.

Procurement Status (Confirmed): As of this review, the team has established formal communication with an authorised UKRA-approved vendor (**Wizards Rockets**). The order for the primary **Aerotech G80T-7 (Single Use)** motor has been **placed and confirmed.**

The logistical plan for collection/delivery upon the team's arrival in the United Kingdom is already in place. This proactive procurement ensures that the flight hardware will be available for final integration and safety inspections at the launch site without delays.

2.4.3 Contingency and Back-up Plans

To ensure mission success in the event of primary motor failure or range-specific requirements, a backup plan is maintained:

- **Designated Backup:** Aerotech G79W-7 (Single Use).
- **RSO Feedback Compliance:** Strictly following the Range Safety Officer's (RSO) PDR recommendations, the team has mandated the **Single Use (SU)** variant for the backup motors. This avoids the logistical complexity of proprietary aluminium reloadable casings (RMS) and ensures 100% mechanical compatibility with the vehicle's 29 mm motor mounting system under any launch-day scenario.

2.5 Recovery Subsystem

The recovery subsystem is designed around a passive, pyrotechnic-driven single-event deployment. At the designated delay time, the motor's ejection charge will pressurise the airframe, shearing the friction fit of the nosecone and simultaneously deploying both the Launch Vehicle's (LV) independent parachute and the CanSat payload with its own independent parachute.

1) Ejection Systems

a. Timings: The deployment relies on the internal delay grain of the COTS Aerotech G80T-7 motor. The fixed 7-second delay initiates immediately post-burnout (1.71 s). Consequently, the ejection event occurs at exactly **8.71 seconds** into the flight. Simulation confirms this occurs at an altitude of approximately 465 m, at a velocity of 12.1 m/s (shortly prior to the 476 m apogee).

b. Electronics: The launch vehicle intentionally **does not utilise ejection electronics or pyrotechnic altimeters**. By relying on the motor's integral pyrotechnic delay charge, the system eliminates the risks associated with software deployment logic, wiring faults, or battery failure during the high-vibration ascent phase. This purely mechanical/pyrotechnic approach maximises mission assurance for the primary recovery event. This directly addresses reliability concerns raised during the PDR phase regarding system robustness

c. Amount of Black Powder Estimate: No raw black powder will be handled or estimated by the team. The ejection energy is provided entirely by the **factory-sealed ejection charge** integral to the Aerotech G80T single-use motor (nominally ~1.0 to 1.5 grams of standard 4FFFF propellant). This strict reliance on COTS hardware guarantees consistency and complies with maximum safety standards on the launch range. This ensures consistency and repeatability, eliminates handling risks, and complies with safety regulations.

2) Descent Systems

a. Description and Sizing of Chutes: To prevent entanglement, the Launch Vehicle and the CanSat descend on completely independent parachutes, separated dynamically by the ejection charge.

- **Launch Vehicle Chute:** Top Flight Recovery PAR-18. A 45.7 cm (18-inch, Area of $0.166\ m^2$) diameter flat circular parachute manufactured from 1.7 oz ripstop nylon. It utilizes 6 shroud lines and is tethered to the airframe via a 14 mm Kevlar shock cord to withstand the 12.1 m/s opening shock. Preventing structural failure or zippering.

- **CanSat Chute:** A separate hexagonal parachute designed specifically for the 350 g payload. The canopy area is strictly calibrated to $0.14\ m^2$ (assuming a nominal C_D of 1.75) to balance safe impact energy against wind drift limitations.

b. Calculations for Descent Rate and Drift: Drift analysis was conducted under the UKRA's maximum permitted launch wind profile of 20 mph (8.94 m/s).

- **Launch Vehicle:** The post-burnout airframe mass (~600 g) descending under the PAR-18 chute yields a descent rate of **12.0 m/s**. Total descent time from 476 m is approximately 40-45 seconds.
- **CanSat Payload:** The 350 g CanSat descending under its specific chute yields a controlled descent rate of **7.0 m/s**. Total descent time from 476 m is approximately 68 seconds. The CanSat mass used in OpenRocket simulations corresponds to an equivalent payload model and may differ slightly from the final integrated mass. This assumption is used exclusively for aerodynamic simulation purposes and is further analyzed in Section 3.2.
- **Drift analysis:** Using wind speed and the descent time, for worst-case when speed wind is 9 m/s, for the launch vehicle, with 44 seconds, the drift would be approximately **396 m**, remaining within the safe recovery radius (~400 m). For the CanSat, with a descent time of 70 seconds, the drift would be approximately **630 m**, remaining within the safe recovery radius (~750 m requirement). The system satisfies competition constraints; however, wind remains the dominant factor affecting landing dispersion.

3) Flight Computer

a. Description of the Flight Computer:

- As the primary recovery system is pyrotechnically driven by the motor, **the Launch Vehicle does not house a flight computer**. The Flight Computer is exclusively housed within the separate CanSat module and operates strictly as a telemetry and data-logging payload. The architecture centres on an ESP32 microcontroller, interfacing with a BMP180 barometric sensor for altitude estimation, an MPU6050 IMU for 6-DOF kinematic logging, and a LoRa E32 transceiver (configured strictly to 25 mW to comply with Ofcom IR 2030 regulations) for live ground station downlink.

b. Verification Plan

- As stated, the CanSat will be the only module housing electronics so, to ensure reliability, it will undergo ground functional testing, sensor calibration, telemetry validation, vibration testing, simulated launch conditions, drop testing, validation of data logging during descent, end-to-end system testing, and full mission simulation including telemetry and recovery. This addresses previous feedback regarding insufficient validation and ensures system readiness prior to launch.

2.6 Ground Control System (GCS) Design

Our Ground Control System (GCS) has been designed as a robust, redundant receiving and monitoring platform capable of acquiring telemetry from the CanSat in real time, processing it locally, storing it persistently, and making it available remotely via a web dashboard. The diagram of the ground station is shown below, in Figure 4.

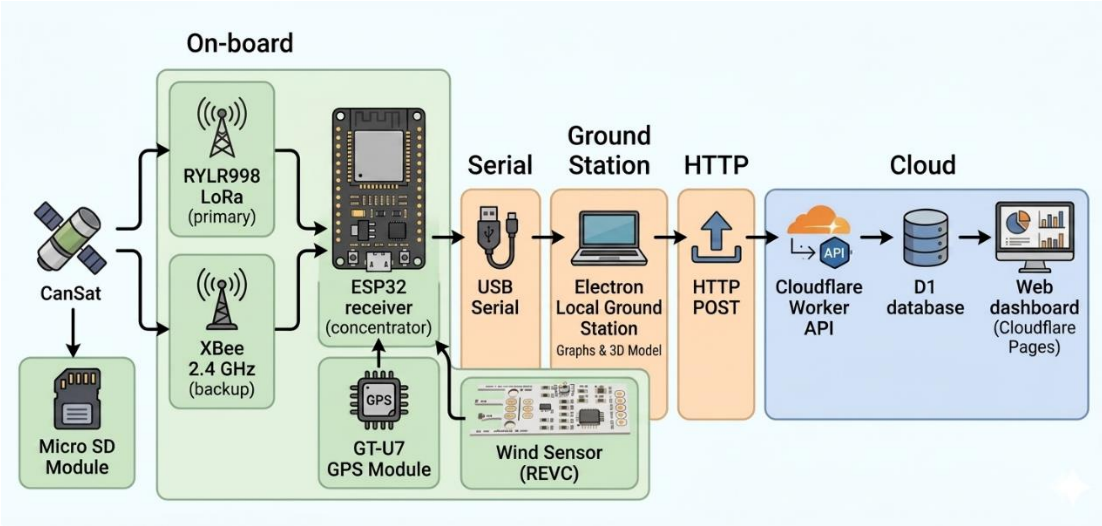


Fig 4. Ground Station diagram

Fig. 4. Diagram of the ground station

The receiving side consists of:

- Two radio modules on the CanSat (RYLR998 LoRa and XB3-24AUT-J XBee) that transmit independently.
- On the ground, an ESP32 microcontroller acts as a radio concentrator where it receives packets from both LoRa and XBee modules through UART serial communication and forwards all data through a single USB serial port to the laptop running the ground station software.
- The local ground station is an Electron desktop application that parses, processes, displays and forwards the telemetry.
- Finally, the Electron app sends the normalised data asynchronously via HTTP to a Cloudflare Worker, which stores it in a Cloudflare D1 SQLite database and serves it to the web dashboard.
- Additionally, within the CanSat payload, there is an SD card module that stores data for later processing once the mission is completed.
- The ground station incorporates a GT-U7 module that allows the implementation of the real-time cargo tracking and recovery system. The Rev.C wind sensor is implemented at the ground station in order to obtain relevant wind information to perform drift calculations.

a) Primary communication interface

The primary communication link is based on IR2030/1/19 interface from IR 2030 – UK Interface Requirements 2030, where it establishes a frequency band of 869.40 - 869.65

MHz, a maximum transmit power of 500 mW e.r.p. and a duty cycle limit of ≤10% (REF-1).

The module chosen for this communication interface is the RYLR998, based on a NUVOTON MCU combined with a Semtech LoRa chipset and operates in the standard 868/915 MHz ISM bands, with a frequency range of 820–960 MHz and a precision of ±10 ppm. Transmit power is configurable from 0 to 22 dBm (approximately 158.48 mW); however, to comply with CE RED certification (EU), the maximum transmission power is limited to 14 dBm using the command AT+CRFOP=14. Communication is performed via a UART serial interface with built-in firmware that supports standard AT commands (REF-2).

For its operation, the center band of 869.5 MHz was established. Considering that the module's band accuracy ranges from ±10 ppm (REF-2), the bandwidth is reflected as 10 ppm = 8.695 kHz, resulting bandwidth between 869.491305 MHz - 869.508695 MHz ≈ 869.49-869.51 MHz.

This falls within the interface range of 869.40 – 869.65 MHz. The frequency is set by code using the following command:

```
lora_send_command("AT+BAND=869500000");
```

Additionally, there is a maximum operating power (ERP) of 500 mW, but since CE certification only applies to this device with a maximum power of 14 dBm (REF-2), this maximum limit is used. The gain of the integrated helical antenna is unknown, so it is assumed to be between + 1.5 dBi and + 2 dBi (based on similar antennas of that type).

The output power, considering an antenna gain of + 2 dBi, should reach a maximum of 14 dBm (25.12 mW), thus satisfactorily meeting the maximum transmission power requirement. The transmission power is set by code using the following command:

```
    lora_send_command("AT+CRFOP=12");
```

For more precise monitoring, RF power measurement tests will be conducted using specialized equipment to adjust the LoRa module's output power, considering the compensation provided by the integrated antenna. CE certification compliance is a primary objective for the team, but, if necessary, it can be waived to increase transmission power and ensure a stable and long-range connection, while maintaining a power margin well below the total 500 mW e.r.p.

Regarding the work cycle, the following is proposed. A set payload will be sent with the 14 variables of the CanSat sensors, declared in types int16_t (2 bytes) and int32_t (4 bytes), and according to the length and precision with which it is required to work, there is a total of 32 bytes to send.

```
typedef struct {
    uint16_t pressure;
    int16_t  temperature;
    int16_t  accel[3];
```

```c
    int16_t  gyro[3];
    int16_t  mag[3];
    int16_t  altitude;
    int32_t  latitude;
    int32_t  longitude;
} cansat_t;
```

Using the parameters for a binary payload with a maximum size of 53 bytes, an SF9, a bandwidth of 125 kHz, and a DR of 1 are obtained (REF-3). For defining these parameters within the command lines, the manufacturer's conventions are used: 9 for the SF9 spreading factor, 7 for the bandwidth of 125 kHz, 1 for a default coding rate of 4/5, and an additional preamble variable with a default value of 12 (REF-4). To adjust the radio frequency power, the following command is used (REF-4):

AT+PARAMETER=<Spreading Factor>, <Bandwidth>, <Coding Rate>, <Preamble>

Adjusted in the code, it is implemented on the following line:

```c
    lora_send_command("AT+PARAMETER=9,7,1,12");
```

To obtain a duty cycle ≤10%, the time rate (T_s) must first be obtained using the symbol rate (R_s) from the following formula (REF-5).

$$T_s = \frac{1}{R_s} \quad ; \quad R_s = \frac{BW}{2^{SF}}$$

Substituting, we obtain:

$$T_s = \frac{1}{\frac{125\ kHz}{2^9}} = 4.096\ ms$$

The packet's duration in the air is the sum of the time rate and the preamble time $\left(T_{preamble}\right)$. The formula for preamble time is as follows (REF-5).

$$T_{preamble} = \left(n_{preamble} + 4.25\right)T_s$$

Substituting the values into the formula:

$$T_{preamble} = (12 + 4.25)4.096\ ms = 66.56\ ms$$

The payload duration ($T_{payload}$) is the number of symbols in the payload ($n_{payload}$) multiplied by the time rate, as expressed in the following formula (REF-5):

$$T_{payload} = n_{payload} \times T_s$$

The number of symbols in the payload ($n_{payload}$) is obtained from the following formula (REF-5):

$$n_{payload} = 8 + \left(\frac{8PL - 4SF + 28 + 16CRC - 20IH}{4(SF - 2DE)}\right)(CR + 4)$$

where:

- PL is the number of payload bytes (from 1 to 255)
- SF is the spread factor (from 6 to 12)
- IH = 0 when the header is enabled, IH = 1 when no header is present
- DE = 1 when LowDataRateOptimize = 1, DE = 0 otherwise
- CR is the encoding rate (1 corresponds to 4/5, from 4 to 4/8)

Substituting, we obtain the following number of payload symbols:

$$n_{payload} = 8 + \left(\frac{8(32) - 4(9) + 28 + 16(1) - 20(0)}{4\big(9 - 2(0)\big)}\right)(1 + 4) = 44.66 \approx 45$$

Therefore, the payload duration is:

$$T_{payload} = 45 \times 4.096\ ms = 184.32\ ms$$

Finally, from the payload duration and the preamble time, we obtain the on-air time (REF-5):

$$T_{on-air} = T_{preamble} + T_{payload}$$

Substituting, we obtain that the on-air time (T_{on-air}) is:

$$T_{on-air} = 66.56\ ms + 184.32\ ms = 250.88\ ms \approx 251\ ms$$

With a Given an on-air time of 251 ms, the transmission period required to achieve a duty cycle of less than 10% is calculated.

$$Duty\ cycle(\%) = \frac{T_{on-air}}{T_{period}} \times 100$$

$$T_{period} = \frac{T_{on-air}}{Duty\ cycle(\%)} \times 100$$

Substituting the values, we obtain T_{period}:

$$T_{period} = \frac{0.251\ s}{10} \times 100 = 2.51\ s$$

To achieve a 10% duty cycle, the payload must be transmitted for a minimum of 2.51 s. Setting the parameter to 3 seconds, we get a duty cycle of:

$$Duty\ cycle(\%) = \frac{0.251\ s}{3\ s} \times 100 = 8.3666\% \approx 8.37\%$$

The duty cycle would be 8.37%, satisfactorily meeting the maximum margin of 10%. In the code, the line responsible for defining the payload sending time is the following:

```
        vTaskDelay(pdMS_TO_TICKS(3000));
```

This function is located within the while loop (1), so it executes continuously every 3 seconds after the last data transmission, ensuring that the system properly fulfills the duty cycle.

NOTE: These duty cycle adjustment parameters are provisional, as they are still in experimental stages to achieve a longer payload transmission time during operation (shorter time between transmissions). In any case, the above serves as a declaration that a fully functional and current standard for the IR2030/1/19 interface parameters has been defined.

b) Secondary communication interface

The secondary communication link is based on IR2030/1/22 interface from IR 2030 – UK Interface Requirements 2030, where it establishes a frequency band of 2400 – 2483.5 MHz and a maximum transmit power of 10 mW e.i.r.p. (REF-1).

The module chosen for this communication interface is the Digi XBee 3 802.15.4, model XB3-24AUT-J, built around a Silicon Labs EFR32MG SoC and operates in the 2.4 GHz ISM band using the IEEE 802.15.4 protocol and features a U.FL antenna connector. The module offers a line-of-sight range of up to 1200 m (4000 ft) outdoors with a maximum transmission power of +8 dBm (adjustable to lower values). It supports multiple serial interfaces (UART, SPI, and I2C). The standard (non-PRO) version provides full regulatory documentation for both CE (EU) and UKCA (United Kingdom) certification (REF-6).

The center band frequency is set by sending an AT command from the microcontroller to the XBee module. Each command begins with AT followed by two letters representing the command to be configured, space, and the parameter in HEX format (REF-7).

According to the Digi XBee 3 ZigBee documentation, the parameters for configuring the frequency band (CH) are set using its HEX commands (REF-7). It was decided to work with the 2.410 GHz frequency, therefore the appropriate channel to implement is 0C (REF-7). The resulting AT command is:

ATCH 0C

In the code it is implemented as follows:

```
    send_cmd("ATCH 0C");
```

Similar to bandwidth, this command is configured using the transmission power (PL) and an integer parameter that determines the decibel level (REF-7).
For 8 dBm (approximately 6.31 mW), parameter 4 is used (REF-7), resulting in the following AT command:

ATPL 4

In the code, it is implemented as follows:

```
    send_cmd("ATPL 4");
```

Regarding the antenna, this model has a U.FL connector, which allows for the use of an antenna with a gain of +1.5 dBi to +2 dBi to achieve the 10 mW (10 dBm) limit imposed by the interface. The ANTX100P001B24003 (PCB antenna) and a reverse-polarization dipole antenna, both with a gain of 2 dBi, are suggested. The final choice will depend on which option provides a stable telemetry connection over the longest distance.

Fig. 5. Antenna

There are no restrictions of any kind for this communication interface regarding duty cycles; however, the same 3-second delay will be applied to payload transmission using the LoRa protocol in order to standardize the data transmission process. This is subject to change based on potential considerations in future testing.

In terms of operating time, the ground station is expected to function for the full launch window using portable battery power. The radio receivers have relatively low power demand, while the laptop can typically operate for several hours on its internal battery. Under nominal operation, the complete GCS is expected to remain functional to remain functional for at least 3 to 5 hours, which is sufficient for pre-launch checks, flight operations, and immediate post-landing recovery. As an additional safety measure, an external power bank will be available to extend laptop and receiver operation if delays occur. Therefore, no fixed external power source is required, although portable backup power is recommended.

Telemetry will be displayed in real time on a laptop-based interface connected to the ground receivers through serial communication. The interface will show the most relevant flight parameters, including altitude, GPS position, temperature, pressure, and mission status flags. In addition to real-time visualisation, all received telemetry packets will be automatically recorded on the ground station laptop in CSV or text-log format for post-flight analysis. This ground logging is complemented by onboard SD card data storage to ensure data recovery even if radio communication is temporarily lost. Furthermore, as an emergency measure, we have a web-based ground station, where the physical ground station sends data to an API designed to receive and display the data online.

No licenses are required to transmit on these frequencies. All transmissions operate under the UK Interface Requirement IR2030 for non-specific short-range devices (airborne permitted). On the contrary, the module manufacturers offer certification guarantees:

- RYLR998: CE RED (UE), under 14 dBm operation (REF-2).
- XB3-24AUT-J: CE (UE) and UKCA (United Kingdom) (REF-8, REF-9).

To ensure operation in Machrihanish environmental conditions, the GCS will be prepared for wind and light rain exposure. The laptop and receivers will be placed inside protective enclosures or covered operating stations, and antennas will be mounted securely to prevent loss of alignment or mechanical instability in strong wind. Waterproof covers, cable strain relief, and weather-protected electronics housings will be used to reduce the risk of rain-related failures. These measures are intended to maintain telemetry reception reliability under the expected field conditions.

2.7 Launch Vehicle Requirements Verification Matrix

Functional Requirements

ID	Description	Justification	Verification Method
FUN- 1	The vehicle must reliably separate and deploy independent recovery systems for the airframe and CanSat.	Ensures safe deceleration and completely prevents entanglement between the two descending masses.	**Demonstration & Test** (Ground-based ejection test)
FUN -2	The vehicle must positively retain the primary motor throughout the entire flight profile.	Prevents the 102 N peak thrust from driving the motor forward into the avionics, and prevents rearward ejection during the delay charge.	**Inspection & Demonstration** (Mechanical fit check)

Performance Requirements

ID	Description	Justification	Verification Method
PER-1	The vehicle must achieve an off-rail velocity of at least 13.0 m/s upon leaving a 2.0 m guide rail.	Ensures sufficient aerodynamic authority for the fins to prevent weathercocking in up to 20 mph crosswinds.	**Analysis** (6-DOF OpenRocket Simulation)
PER-2	The descent drift of all vehicle components must not exceed 750 m under maximum permitted wind conditions.	Strict adherence to Mach-26 range safety limitations and UKRA guidelines for a 476 m apogee.	**Analysis** (Kinematic descent calculation)
PER- 3	The airframe and CanSat must land with a descent velocity not exceeding 12.0 m/s and 7.0 m/s respectively.	Minimises kinetic energy upon impact to protect onboard electronics and prevent hazardous debris.	**Analysis** (Aerodynamic drag mathematical modelling)
PER-4	The vehicle must reach an apogee within ±10% of the 450 m target.	Ensures mission performance accuracy.	**Analysis** (OpenRocket simulation)

Design Requirements

ID	Description	Justification	Verification Method
DES-1	The launch vehicle must withstand aerodynamic forces during ascent.	The nosecone is designed with a 5.0 mm solid PETG wall thickness, preventing deformation under stagnation pressure.	**Analysis** (OpenRocket mass/density calculation) **& Inspection**
DES-2	The primary airframe and all aerodynamic surfaces must be constructed exclusively from non-metallic materials.	Mandatory Mach-26 safety rule to minimise hazard severity in the event of a ballistic trajectory or impact.	**Inspection** (Visual material validation)
DES-3	The fins must possess sufficient shear modulus to withstand aeroelastic flutter at a peak velocity of 102 m/s.	The fins utilise an in-house, cross-grain quasi-isotropic cedar laminate cured under uniform mechanical pressure.	**Analysis & Test** (3-point bend stress test)
DES-4	The nosecone must have a minimum wall thickness of 5 mm PETG.	Prevents deformation under aerodynamic pressure.	**Inspection**

Interface Requirements

ID	Description	Justification	Verification Method
INT-1	The nosecone shoulder must maintain a friction fit during ascent but shear reliably under pyrotechnic pressurisation.	Prevents premature drag-separation during the 1.71 s motor burn while ensuring the parachutes deploy at the 8.71 s mark.	**Test** (Shear pin / Friction fit ground test)
INT-2	The airframe must interface seamlessly with a standard competition launch rail.	Ensures a zero-degree angle of attack until safe off-rail velocity is achieved.	**Demonstration** (Rail button sliding fit)
INT-3	The 14 mm Kevlar shock cord must interface securely with the primary airframe bulkheads without shearing the Kraft tubing.	The Kevlar must absorb the kinetic energy of the early 12.1 m/s parachute deployment.	**Analysis & Inspection** (Tensile load calculation)

Operational Requirements

ID	Description	Justification	Verification Method
OP-1	The CanSat avionics must feature an external mechanical switch to be	Protects range personnel from electrical hazards and preserves battery life during pre-launch range delays.	Demonstration

	safed or armed while on the launch pad.		
OP-2	The Ground Control Station (GCS) must operate continuously on battery power for the duration of the launch window.	Ensures telemetry recording is not interrupted during extended 3-to-5-hour pad delays without relying on mains power	**Test** (Battery longevity run)

Additional requirements

ID	Description	Justification	Verification Method
ADD-1	RF Transmission must strictly comply with UK Ofcom IR 2030 regulations for Short Range Devices (SRD)	Legal requirement. The LoRa module is software-attenuated to ≤ 25 mW within the 868 MHz band.	**Inspection** (Firmware code review)
ADD-2	Solid rocket motors must be legally procured within the UK.	International aviation law prohibits transporting propellants from Mexico. Motors are sourced via UKRA-approved vendors.	**Inspection** (Purchase order validation)

3. CanSat payload

3.1 Payload Final Design

3.1.1 Overall Structural Design

The final CanSat structure consists of a main cylindrical enclosure with an outer protective shell that covers and secures all internal components, as shown in Fig. 6. The system has an overall height of 18 cm and a maximum diameter of 7 cm, corresponding to the external cylindrical housing. This outer shell provides mechanical protection to the payload during handling, launch, deployment, and descent, while also defining the main geometric limits of the CanSat.

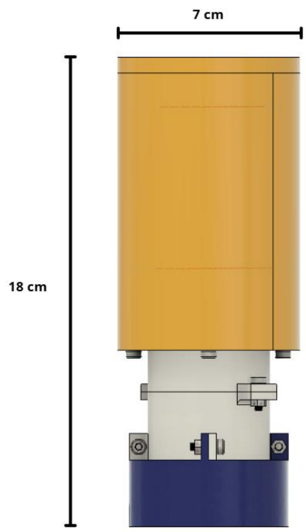


Fig. 6. Front view of the CanSat with overall dimensions

The internal architecture is organized vertically to maximize space efficiency and maintain stable component positioning, as illustrated in Fig. 7. The electronics are mounted inside the upper section of the structure, while the battery is placed in a dedicated lower space to optimize the distribution of mass and improve assembly accessibility. The egg module is located in the central region of the system, where it is mechanically supported and isolated from direct external impact. This arrangement ensures that each subsystem has a specific location within the structure, reducing interference between components and improving overall functionality.

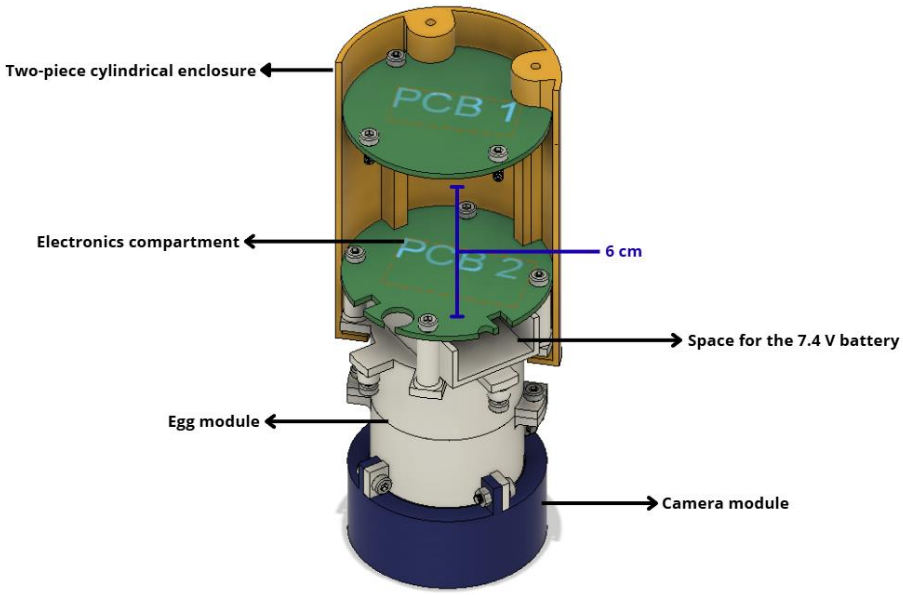


Fig. 7. Internal structural layout of the CanSat showing PCB arrangement.

At the bottom of the CanSat, the camera module is integrated with a downward-facing camera aperture, allowing the camera to record the lower side of the vehicle during descent, as shown in Fig. 8. This configuration ensures unobstructed visual data acquisition while maintaining structural integrity in the lower section.

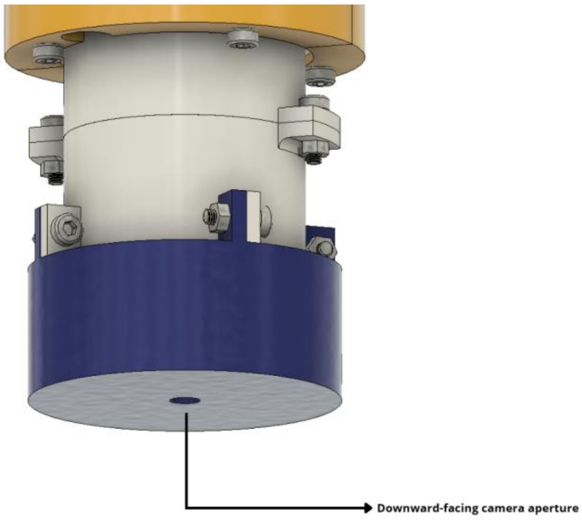


Fig. 8. Bottom view showing the downward-facing camera aperture.

The enclosure is assembled using M3 screws and, in selected locations, nuts, which provide a secure and repeatable mechanical connection between the different structural parts. This fastening method allows the CanSat to remain rigid during flight while also enabling disassembly for maintenance, inspection, and testing.

In addition, the top section includes a parachute attachment point integrated into the top end cap, which is essential for a reliable deployment subsystem, as shown in Fig. 9. This attachment point ensures that the parachute can be firmly connected to the CanSat and that the ejection and descent phases occur in a controlled and stable manner.

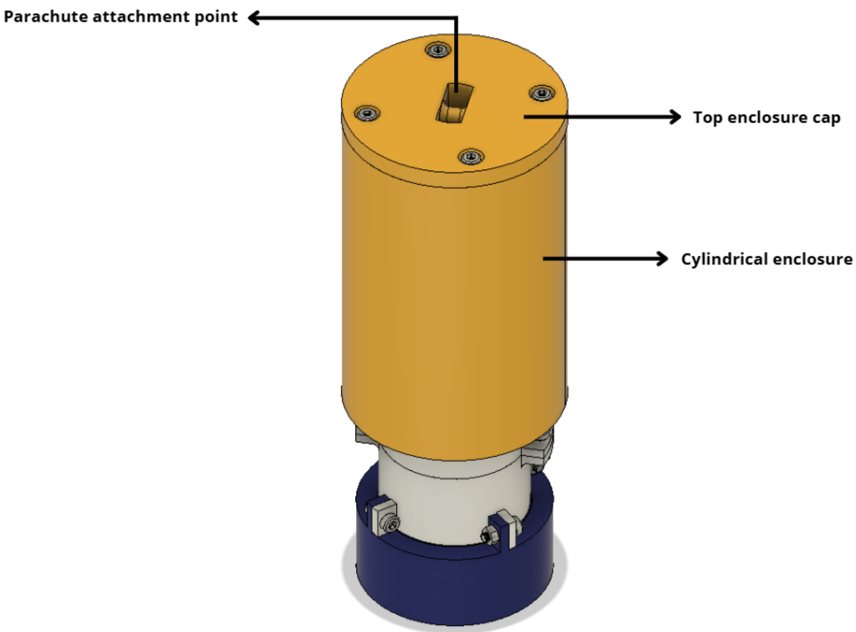


Fig. 9. External view of the cylindrical enclosure and top end cap.

3.1.2 Structural Components

The structural design of the CanSat consists of a reduced yet efficient set of components that ensure mechanical integrity while maintaining a lightweight and compact configuration.

The primary structural element is the cylindrical enclosure, which functions as the main load-bearing structure and provides protection for all internal subsystems. Its geometry allows for an efficient distribution of mechanical stresses during launch, ejection, and descent.

The top end cap serves as the upper closure of the system and incorporates the parachute attachment point, enabling the transmission of forces generated during parachute deployment. This component plays a critical role in maintaining structural stability throughout the descent phase.

At the lower end, the bottom section provides structural closure while integrating a downward-facing camera aperture, allowing sensor operation without compromising the rigidity of the enclosure.

In addition to the external structure, the enclosure integrates internal support columns (standoffs) that are directly incorporated into the body of the structure. These elements are used to mechanically support and fix the PCBs in position. The upper PCB is mounted onto these columns using M3 screws, while additional columns are positioned on top of the egg module to support the lower PCB. This configuration ensures proper alignment, prevents unwanted movement, and contributes to the overall structural stability of the system. The arrangement of these internal supports and their interaction with the PCBs is illustrated in Fig. 10.

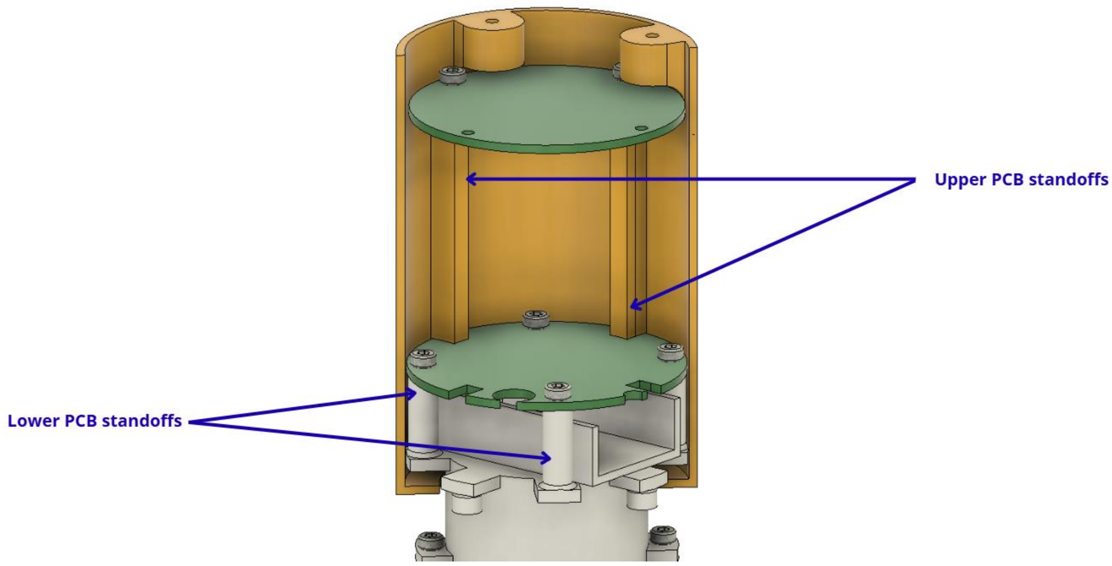


Fig. 10. Standoff configuration for mounting the upper and lower PCBs.

All structural elements are assembled using a fastening system based on M3 screws and nuts, which provides a secure and reliable mechanical connection. This system ensures sufficient rigidity to withstand expected mechanical loads while also allowing straightforward assembly, disassembly, and maintenance operations.

3.1.3 Changes since PDR

Several modifications were implemented in the final design of the CanSat to improve environmental protection, structural integrity, and overall system reliability compared to the Preliminary Design Review (PDR).

In the initial design, the cylindrical enclosure included multiple lateral openings and perforations, particularly in the lower section and along the side walls. Additionally, the upper section of the CanSat did not incorporate a top end cap, leaving internal components partially exposed to the external environment. While this configuration allowed accessibility and reduced material usage, it introduced potential risks related to environmental exposure and mechanical vulnerability.

In the final design, these openings were eliminated, resulting in a fully enclosed cylindrical structure. The enclosure now incorporates a top end cap, as shown in Fig. 9, effectively sealing the upper section that contains the main electronics. This modification was implemented to protect sensitive components from external factors such as moisture, dust, and potential contaminants, which are particularly relevant given the expected environmental conditions during launch operations.

Similarly, the lower section of the CanSat was redesigned. In the PDR configuration, this region included perforations and served as a housing for an additional PCB. In the final design, the number of PCBs was reduced from three to two, simplifying the internal architecture and improving space efficiency. As a result, the lower module was repurposed into a dedicated camera compartment, now featuring a downward-facing camera aperture (see Fig. 8). This aperture is minimized to the required size and is effectively sealed by the camera lens itself, preserving the protective integrity of the enclosure.

Another significant modification involves the relocation of the battery. In the PDR design, the battery was positioned below the egg module, directly above the third PCB. In the final configuration, the battery was moved to a position above the egg module. This change was implemented to improve the mass distribution of the system and contribute to a more balanced internal layout. By relocating the battery, the design reduces potential interference with lower subsystems and improves accessibility during assembly.

These changes collectively enhance the robustness of the structure by minimizing direct exposure of internal components to the external environment. The fully enclosed configuration reduces the risk of damage due to impact during landing and prevents the ingress of water or debris, which is critical for both electronic reliability and battery safety.

Overall, the transition from a partially open design to a sealed enclosure, along with the optimization of internal component distribution, represents a significant improvement in terms of mechanical protection, environmental resistance, and system reliability, while maintaining the functional requirements of the CanSat mission.

3.1.4 Additional mission: quail egg cushioning

Building on the previous implementation of this mission, a cushioning system based on expanded polystyrene spheres has been designed to absorb the vibrations and impacts that the egg might experience throughout the launch and flight process.

Fig. 11. Cushioning system for the quail egg

As can be seen in Figure 11, the complete system consists of two **polystyrene foam** lids (shown in grey), an inner filling of **expanded polystyrene beads** for shock absorption (modelled as a white cylinder) and a moulded thermoplastic polyurethane (TPU, shown in blue) shell to contain the polystyrene beads. The main advantage of this design is that it does not depend on the egg having a specific shape or size, as the beads adapt to the available space within the designated area thanks to their small size.

The outer TPU mould serves to contain the expanded polystyrene spheres whilst dampening any minor vibrations the system may experience, thanks to the material's inherent elastic properties; the polystyrene foam caps act as shock absorbers on the z-axis, as this is where the launch and decoupling forces are concentrated, whilst the expanded polystyrene spheres, due to their small diameter (~2–2.5 mm), act as a granular fluid, dispersing forces and vibrations whilst preventing rigid contact points with the egg.

3.1.5 Estructural integrity and vibration mitigation

One of the main considerations in the structural design of the CanSat is the potential loosening of threaded fasteners due to vibrations experienced during launch and ejection. Since the assembly relies on M3 screws and nuts to secure both the external enclosure and internal components, maintaining the integrity of these connections is critical.

The CanSat structure is based on a two-piece cylindrical enclosure with a top end cap, as shown in Fig. 7, where all structural elements are mechanically joined using threaded fasteners. Vibrational loads generated during launch could induce relative motion between components, potentially leading to gradual loosening if not properly addressed.

To mitigate this risk, all fasteners are securely tightened, and in critical joints, nuts are used in combination with screws to increase resistance against vibration-induced loosening. In addition, the compact geometry of the structure (18 cm height and 7 cm diameter) reduces the magnitude of inertial forces acting on the system, contributing to overall mechanical stability.

Furthermore, internal components are not solely dependent on fasteners for stability. As illustrated in Fig. 8, the enclosure integrates internal support columns (standoffs) that provide direct mechanical support for the PCBs. The upper PCB is fixed onto these standoffs using M3 screws, while the lower PCB is also constrained by dedicated mounting points. This configuration prevents relative displacement between components and reduces mechanical stress on the fasteners.

In addition to the design measures described above, the assembled CanSat will be subjected to controlled vibration testing using a dedicated vibration system available to the team. This testing phase is intended to evaluate the mechanical stability of the fully assembled structure under simulated vibration conditions. The objective is to identify any potential weaknesses, such as fastener loosening or structural instability, and implement corrective actions if necessary prior to launch.

Through the combination of a robust fastening strategy and planned experimental validation, the system is expected to maintain its structural integrity under operational conditions, minimizing the risk of mechanical failure during flight.

3.1.6 Technical justification of manufacturing materials (3D print)

The operating environment of a CanSat inside a rocket imposes three main mechanical challenges: axial acceleration during launch, harmonic vibration from the motor, and the opening shock of the parachute deployment. The materials considered are: Sunlu Nylon **PA12-CF**, Bambu Lab **Polycarbonate (PC)**, and SainSmart Nylon **PA6-CF**.

Table 1 analyzes the data extracted from the manufacturers' Technical Data Sheets (except for the SainSmart PA6-CF, where values are nominal approximations for the material type):

Propety	PA12-CF (Sunlu)	PC (Bambu Lab)	PA6-CF (SainSmart)
Density (g/cm^3)	1.04	1.20	≈1.15
Tensile Strength (X-Y) (X-Y) (MPa)	100 MPa	62±2 MPa	≈110
Flexural Modulus (X-Y) (MPa)	5996	2310±70	≈5500
Elongation at Break (X-Y) (%)	12%	3.8±0.3%	≈4%
Impact Strength (X-Y) (kJ/m^2)	13	34.8±2.1	≈10
Heat Deflection Temp (°C)	176 °C	112 °C	≈160 °C

Table 1. Material Comparison (PA12 vs PC vs PA6)

a. Launch Phase

During ascent, the structure must be rigid enough to avoid entering into resonance with the rocket's vibrations.

- **PA12-CF:** Possesses the highest flexural modulus, ensuring superior structural stiffness. This minimizes the deflection of the CanSat walls, protecting internal sensors from potential mechanical contact.
- **PC:** Having a much lower modulus of elasticity, it is a more flexible material, which could allow for higher amplitude vibrations if the wall design is thin-a common trade-off to reduce weight.

b. Ejection and Deployment Phase

Now the parachute deploys, the structure experiences a sudden snatch load or jolt.

- **PA12-CF:** Features an elongation at break of 12%, which is significantly higher than the 3.8% of the Polycarbonate. This indicates that PA12-CF can deform slightly in an elastic or plastic manner before failure, better absorbing the energy from the parachute deployment.
- **PC:** Its impact strength is the highest, making it more reliable against brittle fractures from sharp impacts during the descent or landing.

c. Moisture and Weight

PA12-CF is 12.7% lighter than Polycarbonate and 9% lighter than PA6-CF. In a CanSat, this weight saving directly translates into a lower descent rate or the ability to add more payload (sensors/passenger). Furthermore, PA6-CF tends to absorb moisture quickly (hygroscopic), which can compromise the dimensional stability of the part or soften it in humid climates. PA12-CF has much lower water absorption, ensuring that the deployment system's tolerance remains constant throughout its operation.

Material's Justification

1. **PA12-CF (Sunlu):** Primary Option. This material is chosen for its excellent strength-to-weight ratio. It is the only material offering 100 MPa of strength with the lowest density in comparison (1.04 g/cm^3). Its high thermal resistance (176 °C) ensures no deformation will occur even if the rocket experiences significant aerodynamic heating. Carbon Fiber reinforcement is a critical factor in this selection; it serves to significantly increase the material's stiffness and provides the dimensional stability necessary to prevent the warping and shrinkage inherent to pure polyamides.
2. **PC (Bambu Lab):** Selection for Fastening Components. Recommended exclusively for parachute anchorage points. It's extremely tough nature and high impact resistance perform better at preventing bolts from "tearing" out of the structure during parachute opening.
3. **PA6-CF (SainSmart):** While PA6-CF filament possesses high-performance mechanical properties suitable for functional parts and prototypes, it is considered technically inferior to the specialized alternatives due to its higher density and marked hygroscopy compared to PA12-CF, as well as its lower impact resistance compared to PC. However, its inclusion in the design is justified by its immediate availability in the team's inventory, allowing for mechanical validation and fit testing to proceed without additional costs; despite being outperformed in

lightness and toughness, its structural characteristics remain robust for the project's development phases.

3.2 Descent Control

For the preliminary parachute design, the drag force expression is taken as the starting point:

$$D = \frac{1}{2}\rho C_D A v^2$$

where:

- D = drag force (N)
- C_D = drag coefficient
- ρ = air density (kg/m^3)
- A = cross-sectional area (m^2)
- v = velocity of the free-falling object (m/s)

If the drag force is assumed to balance the system weight and, based on the literature, a drag coefficient value for an ideal parachute is adopted, then the quantity of interest is the required cross-sectional area needed to cushion the landing:

$$A = \frac{2D}{\rho C_D v^2}.$$

There are different parachute configurations; among the most representative are circular models and Ram-Air types. The latter, since they are composed of inflatable cells that operate as an aerodynamic wing, provide directional control and horizontal displacement. However, they require greater technical complexity, are more sensitive to partial collapse, and are more expensive, making them less suitable for rigid, uncontrolled payloads such as a CanSat. In contrast, circular parachutes stand out for their simpler construction, lower probability of deployment failure, and more constant descent rate. In addition, the incorporation of a central vent helps reduce oscillations and improve aerodynamic stability, which are fundamental qualities for preserving system integrity and ensuring proper data acquisition during the mission. The parachute is stored folded over the CanSat, inside the rocket's undocking compartment, with the suspension lines arranged in a way that minimizes the risk of entanglement during deployment. Such deployment is expected to occur passively, aided by the free-fall condition to which the system will be exposed once released.

The drag coefficient used in the parachute sizing is 1.75, a value obtained from sources available online for similar configurations. This value is suitable for a preliminary estimate, but importantly, it can vary depending on environmental conditions, such as air density, wind speed, and the characteristics of the parachute material.

3.2.1 Parachute Sizing

The required front surface area was calculated from the drag force equation:

$$A = \frac{2D}{\rho C_D v^2}.$$

Substituting the selected values:

- $D = 3.5$ N,
- $C_D = 1.75$,
- $\rho = 1.2$ kg/m^3,
- $v = 3.5$ m/s,

You get:

$$A = \frac{2 \times 3.5}{1.2 \times 1.75 \times 3.5^2} = 0.27\ \mathrm{m}^2.$$

From the balance between the drag force and the weight of the system, a transverse surface of the parachute was obtained. In this way, the parachute can be roughly modeled like a ring inscribed on a hexagon with external diameter and internal diameter of $A = 0.27\ \mathrm{m}^2,\ D = 55$ cm, $d = 3$ cm.

For the final manufacture, a water-repellent fabric was selected whose properties limit the passage of air through the fibers of the material. This behavior increases the drag coefficient and, consequently, favors an effective decrease in the descent speed.

Once the parachute sizing criterion has been established, it is necessary to estimate how the system will behave during descent and to evaluate the accumulated horizontal displacement to the landing point.

3.2.2 Parachute drift modeling

Ideally, after deployment, the parachute-load system would be expected to perform a predominantly vertical descent (z axis) and with a minimal horizontal component. However, in practice the trajectory deviates due to various factors, including:

- the presence of horizontal wind in the plane xy
- the oscillation/pendulum of the load-lines-bell assembly,
- the geometry of the hood (which modifies the effective area A, drag coefficient C_D and stability),
- and initial deployment conditions (tilt, asymmetries, disturbances).

Drift is defined as the cumulative horizontal displacement of the system during descent, measured between the deployment point and the landing point. In general, it is expressed as follows:

$$\Delta r_{xy} = \int_0^{t_f} \mathbf{v}_{xy}(t)dt \qquad (1)$$

where $\mathbf{v}_{xy}(t)$ is the horizontal speed of the system in a frame fixed to the ground.

To describe the entire motion in three dimensions, it is possible to model both the translation and orientation of the system. In this approach, the following are distinguished: (i) a system of absolute axes (fixed to the environment) convenient for integrating the trajectory $\mathrm{r}(t)$, and (ii) a system of body axes (linked to the parachute/load), useful for describing orientation by means of a composition of orthogonal rotations. This separation is practical because several aerodynamic quantities are naturally formulated in body axes, depending on the instantaneous orientation of the system and the relative flow. Consequently, aerodynamic forces are calculated in body axes and transformed into absolute axes by means of a rotation matrix $R(t)$, to finally use them in the translation equation of the center of mass:

$$m\ddot{r}(t) = \sum F_{abs}(t) \tag{2}$$

For practical purposes of the project, it was established that the horizontal displacement of the system should not exceed 400 m. Accordingly, the following mathematical and physical considerations are adopted:

- The aerodynamic effect of the payload is neglected due to its small size; only their weight is considered.
- Only the stationary descent phase is analyzed, since the deployment phase is of very short duration and its effect on the trajectory is negligible.
- The system is modeled as a single rigid solid.
- To verify that the total drift does not exceed 400 m, the three-dimensional model is omitted. The analysis is restricted to two dimensions in the ZX plane, assuming that the maximum displacement in this plane is representative of any radial direction in space.

To adequately describe the model, Hume and Steel group the equations into two main blocks: (i) an equation that represents the motion around the center of mass and (ii) a set of equations that describes the motion of the center of mass itself.

3.2.3 Movement over the center of mass

This equation is not yet following the position, but the $x(t)$, $y(t)$ orientation of the parachute-load system. Because we interpret the system as a rigid body, the system can fall completely straight with respect to the ground or with an angle of inclination, the most likely case observed is previous throws, causing the system to oscillate.

To study this turn, it is natural to apply the kinetic momentum theorem:

$$\sum M_{CM}^{ext} = \frac{dL_{CM}}{dt} \tag{3}$$

That is:

The sum of external moments with respect to the center of mass is equal to the rate of change of angular momentum with respect to the center of mass.

According to Newton's second law:

$$\frac{dL_{CM}}{dt} = I_{CM}\ddot{\theta} \tag{4}$$

where:

- L_{CM} = angular momentum with respect to the center of mass.
- I_{CM} = moment of inertia of the system with respect to the center of mass.
- $\ddot{\theta}$ = angular acceleration.

For the moment of inertia, we start from the model of 2 masses M at a distance l:

$$I_{CM} = \sum M_x L_x^2 \tag{5}$$

Applying this expression to the system of interest:

$$I_{CM} = M_C l_C^2 + (M_P + M_a) l_p^2 \tag{6}$$

where:

- M_C = mass of the load.
- M_P = mass of the parachute.
- M_a = aerodynamic mass.
- l_C = distance between the center of mass of the charge to the center of mass of the system.
- l_p = distance between the center of mass of the parachute and the center of mass of the system.

3.2.4 Centers of mass

To determine the position of the center of mass of a system formed by two-point masses, the following is used:

$$\vec{R}_{CM} = \frac{1}{M} \sum m_i \vec{r}_i. \tag{7}$$

Applying this definition to known distances from the center of mass:

$$l_C = \frac{L(M_p + M_a)}{M_C + M_p + M_a} \tag{8}$$

$$l_p = \frac{L M_C}{M_C + M_p + M_a} \tag{9}$$

In this way, an expression is obtained in which the moment of inertia depends only on the length of the parachute cords:

$$I_{CM} = \frac{M_p L^2 (M_a + M_C)}{M_C + M_p + M_a} \tag{10}$$

3.2.5 Forces

The three moments that generate the turn are identified below:

- $M_C g l_C \sin\theta$: gravitational moment generated by the rotation in one direction of the system's load.
- $-M_p g l_p \sin\theta$: gravitational moment generated by the rotation in the opposite direction of the mass of the parachute of the system.
- $F_N l_p$: force perpendicular to the axis of symmetry of the parachute caused by the flow of air in the canopy.

By bringing these contributions together, you get:

$$I_{CM}\ddot{\theta} + M_c g l_c \sin\theta - M_p g l_p \sin\theta + F_N l_p = 0 \tag{11}$$

Now, if we divide equation 11 by I_{CM} and make the corresponding reductions, we get:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L\left(1 + \frac{M_C}{M_a}\right)} \sin\theta + \frac{F_N}{L(M_a + M_C)} = 0 \tag{12}$$

In addition, considering that:

$$F_N = C_N \frac{\rho V^2 A}{2} \tag{13}$$

$$F_a = C_a \frac{\rho V^2 A}{2} \tag{14}$$

The final equation takes the following form:

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L\left(1 + \frac{M_P}{M_a}\right)} \sin\theta - \frac{C_N \rho V^2 A}{2L(M_a + M_p)} \tag{15}$$

3.2.6 Center of Mass Movement

Applying the same principle together with Newton's second law, we obtain the translational equations of the center of mass in the X and Z directions:

$$M' \frac{d^2 Z}{dt^2} = Mg - F_A \cos\theta + F_N \sin\theta \tag{16}$$

$$M' \frac{d^2 x}{dt^2} = -F_A \sin\theta - F_N \cos\theta \tag{17}$$

where:

- $M' = M_C + M_P + M_a$.
- $M = M_C + M_P$.

Dividing the expressions by M', and making the corresponding substitutions, we obtain:

$$\frac{d^2 Z}{dt^2} = \frac{Mg}{M'} - C_A \left(\frac{\rho A}{2M'}\right) V^2 \cos\theta + C_N \left(\frac{\rho A}{2M'}\right) V^2 \sin\theta \tag{18}$$

$$\frac{d^2 x}{dt^2} = -C_A \left(\frac{\rho A}{2M'}\right) V^2 \sin\theta - C_N \left(\frac{\rho A}{2M'}\right) V^2 \cos\theta \tag{19}$$

3.2.7 Speeds in the system

The absolute velocity of the rigid solid in space is defined as the sum of the relative linear velocity of the system with respect to the incident flow and the proper rotation:

$$v_{espacial} = v_{relativa} + w \cdot r \tag{20}$$

If this definition is applied to the problem posed:

$$U_x = \frac{dx}{dt} - \left(\frac{LM_C}{M'}\right)\left(\frac{d\theta}{dt}\right)\cos\theta + V_x \tag{21}$$

$$U_z = \frac{dz}{dt} + \left(\frac{LM_C}{M'}\right)\left(\frac{d\theta}{dt}\right)\cos\theta \tag{22}$$

Knowing that the angle θ is formed with the vertical of the axis that passes through the center of mass and α is generated between the air flow and the axis of the system, the following relationship is proposed:

$$a + \theta = \tan^{-1}\left(\frac{U_z}{U_x}\right) \tag{23}$$

3.2.8 System of equations

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L\left(1 + \frac{M_P}{M_a}\right)}\sin\theta - \frac{C_N \rho V^2 A}{2L\left(M_a + M_p\right)} \tag{24}$$

$$\frac{d^2 Z}{dt^2} = \frac{Mg}{M'} - C_A\left(\frac{\rho A}{2M'}\right)V^2\cos\theta + C_N\left(\frac{\rho A}{2M'}\right)V^2\sin\theta \tag{25}$$

$$\frac{d^2 x}{dt^2} = -C_A\left(\frac{\rho A}{2M'}\right)V^2\sin\theta - C_N\left(\frac{\rho A}{2M'}\right)V^2\cos\theta \tag{26}$$

$$U_x = \frac{dx}{dt} - \left(\frac{LM_C}{M'}\right)\left(\frac{d\theta}{dt}\right)\cos\theta + V_x \tag{27}$$

$$U_z = \frac{dz}{dt} + \left(\frac{LM_C}{M'}\right)\left(\frac{d\theta}{dt}\right)\cos\theta \tag{28}$$

$$a + \theta = \tan^{-1}\left(\frac{U_z}{U_x}\right) \tag{29}$$

3.2.9 Numerical simulation of drift

To verify that the CanSat did not exceed the 400 m horizontal drift limit set for the mission, a Python code was developed to numerically solve the proposed system of equations. From this procedure it was possible to simulate the trajectory of the system during descent and estimate the maximum expected horizontal displacement.

3.2.10 Parameters used in the simulation

The numerical simulation was carried out using the following parameters: gravity 9.81 m/s^2, air density ρ = 1.225 kg/m3 , load mass Mc = 0.35 kg, parachute mass Mp = 0.1 kg, aerodynamic mass Ma = 0.1 kg rope length L = 0.8 m, parachute area A = 0.27 m2, drag coefficient CA = 1.75, normal coefficient CN = 0.15, transverse wind speed Vx = 3.77 m/s, initial inclination angle θ = 20°, and initial conditions of position and velocity. These parameters were taken from the mission configuration and typical flight conditions. 600 seconds of descent were simulated with a time step of dt = 0.01 s, and the results were recorded every 10 steps. It should be noted that the value of the speed of the transverse wind was taken directly from the data provided by the meteorological station of the Autonomous University of Yucatan at that time.

The Figure 12 shows the results obtained by the simulation, including the trajectory in the XZ plane, the temporal evolution of the horizontal drift, the angular inclination and the magnitude of the relative velocity of the system.

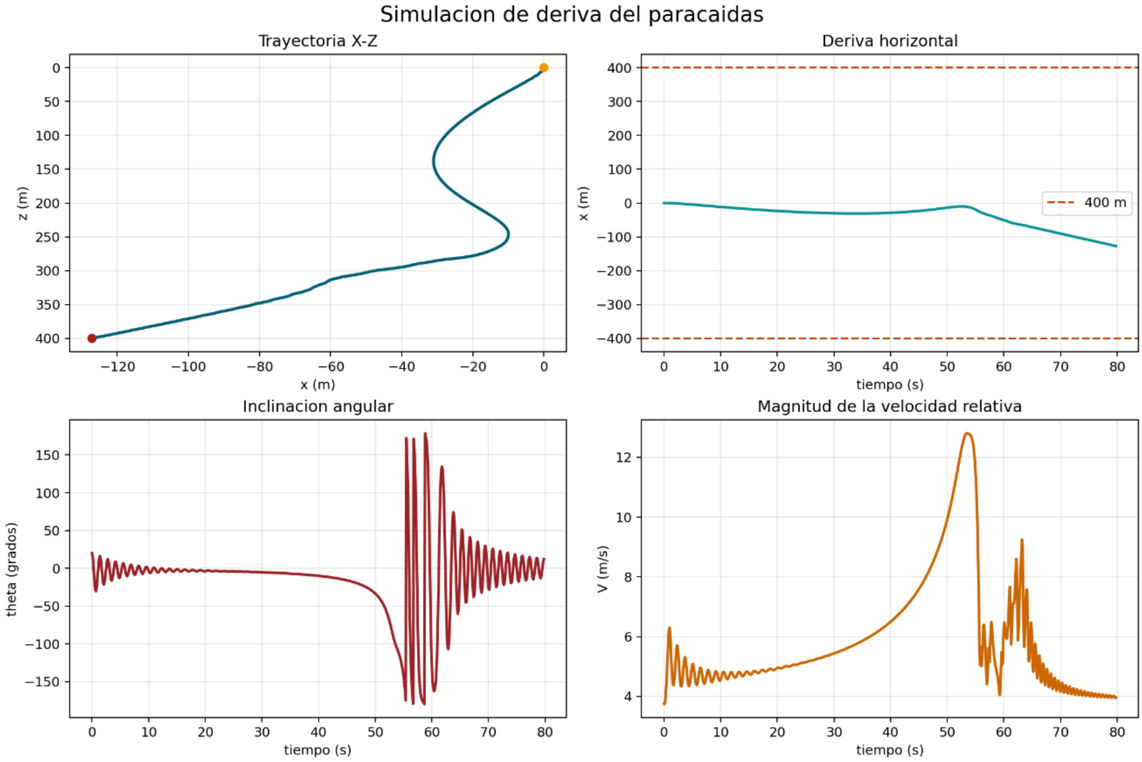


Fig. 12. Parachute drift simulation

Based on these results, a maximum horizontal drift of 127,135 m was obtained, a value that is below the limit of 400 m considered as a safety criterion for the project. Therefore, according to the model used, the system complies with the established horizontal displacement constraint.

However, it is important to note that these results correspond to a preliminary numerical estimate, so it will be necessary to carry out future experimental tests to validate the real behavior of the system and contrast it with the model's predictions.

3.3 Electrical Power Subsystems

A nano-tech model LiPo 2S battery with a 370 mAh capacity was selected for the system due to its high energy density, low weight, and discharge capability. The battery uses a 2S configuration (two cells in series), providing:

- Nominal voltage: 7.4 V
- Maximum voltage: 8.4 V (fully charged)
- Minimum voltage: 6.0 V (to prevent device degradation)

The nominal capacity of the battery is 0.37 Ah, allowing for an estimation of the available energy using the expression:

$$E = V \cdot Ah = 7.4 \cdot 0.37 = 2.74\, Wh$$

Additionally, the battery features a discharge rate of 25–40C, which defines its current supply capacity. The maximum continuous current is calculated as:

$$I_{cont} = 0.37 \cdot 25 = 9.25\, A$$

while the peak current is:

$$I_{peak} = 0.37 \cdot 40 = 14.8\, A$$

These values are considerably higher than the system requirements, as consumption remains below 1 A. This ensures an adequate operating margin and stability against load variations or transient current peaks, especially during wireless transmission events.

Voltage regulation is handled by a step-down DC-DC converter to 3.3 V, which powers most electronic modules, and a 5 V linear regulator dedicated to the GPS module. Power control is implemented via an external switch placed in series with the battery. This switch is positioned to be accessible from the exterior of the CanSat, allowing it to be powered on before launch and safely powered off after recovery.

Consumption was estimated based on the typical operating values of each component (Fig. 13). Table 2 summarizes the main electrical parameters, including operating voltage, average current, power, and duty cycle.

No	Component	Voltage	Current	Power	Duty Cycle
1	ESP32	3.3 V	80 mA	0.27 W	100%
2	LoRa RYLR998	3.3 V	140 mA (300 mA peak)	0.40–1 W	20–30%
3	XBee XB3	3.3 V	135 mA	0.45 W	0–10%
4	GPS	5 V	30 mA	0.15 W	100%
5	GY-87	3.3 V	10 mA	0.03 W	100%
6	microSD	3.3 V	50 mA	0.17 W	10–20%
7	Camera RunCam	5 V	300–500 mA	1.5–2.5 W	100%

Table 2. Energy consumption of system components

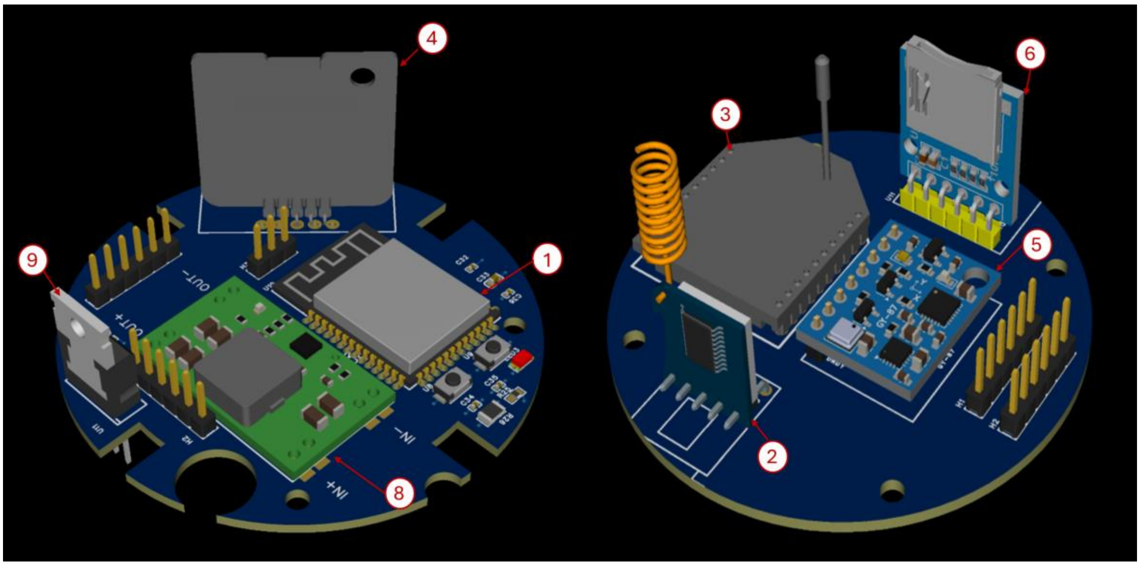


Fig. 13. Left: Lower PCB, Right: Upper PCB

Duty cycles were estimated based on expected system behavior. The microcontroller, sensors, and GPS operate continuously (100%), while the LoRa module transmits data periodically every 1–2 seconds (approx. 20–30% duty cycle). The XBee module is considered a secondary channel with occasional use (0–10%). The microSD card performs intermittent writes (10–20%).

Based on these values, the average consumption of the system (excluding the camera) is estimated at 1.47 W. For a minimum operating time of 45 minutes (0.75 h) the required energy is:

$$E_{\text{without a camera}} = 1.47 \cdot 0.75 = 1.10\, Wh$$

Comparing this to the nominal energy available (2.74 Wh), the system can operate with an adequate margin, meeting the minimum autonomy requirement.

However, including the camera increases total consumption to approximately 3 W, implying an energy demand of:

$$E_{\text{with camera}} = 3 \cdot 0.75 = 2.25\, Wh$$

In this scenario, the operating margin is considerably reduced, approaching the limit of available capacity.

Environmental conditions must also be considered. In Scotland, temperatures are expected between 10°C and 18°C, potentially dropping to 5°C–12°C due to altitude and wind. These conditions cause a reduction in effective battery capacity, estimated at 85% to 95% of its nominal value. Applying this factor:

$$E_{\text{reduced}} = 2.74 \cdot 0.9 = 2.47\, Wh$$

Assuming an overall regulation system efficiency of 85%, the useful energy available is reduced to:

$$E_{usseful} = 2.47 \cdot 0.85 = 2.10\, Wh$$

Under these conditions, the estimated operating time without the camera is:

$$t_{opp} = \frac{2.10}{3} = 0.70h \approx 42\ min$$

This result shows that the system remains very close to the 45-minute requirement, which is encouraging considering that the calculation already includes both environmental derating and power regulation losses. Under nominal conditions, the available energy is sufficient to support the mission duration, while under colder operating conditions the safety margin becomes smaller but still remains reasonably close to the target.

For this reason, and to improve the robustness of the electrical design, an independent power source for the camera is still recommended. This approach reduces the load on the main system, increases operational reliability, and provides additional safety margin for the critical onboard electronics.

3.4 Category Challenge Integration

The payload's primary purpose is to acquire, store, and transmit high-quality telemetry during ascent and descent, complying with Category 1: Launch a rocket with a CanSat as close to 450 m as possible. It incorporates all required standard sensors (altimeter and GPS) and adds an innovative hot-wire anemometer on the Ground Control System. In line with the Mach-26 Guidelines (REF-13), the payload also records video throughout the flight, as a challenge. By combining redundant telemetry, local SD storage, and cloud backup, the system guarantees that all mission data will be recovered and analyzed, directly supporting both scoring and safety objectives.The following components and modules are part of the payload:

- Altimeter: **GY-87 module** (MPU9250 9-DOF IMU + BMP280 barometer). Dimensions: 25 × 14 × 3 mm. Operates at 3–5 V via I2C. Provides pressure (300–1100 hPa, resolution 0.16 Pa), temperature (±1 °C), and 9-DOF inertial data: 3-axis accelerometer with selectable full-scale ranges of ±2 g, ±4 g, ±8 g, and ±16 g; 3-axis gyroscope with selectable full-scale ranges of ±250 °/s, ±500 °/s, ±1000 °/s, and ±2000 °/s; 3-axis magnetometer with a range of ±4800 µT (REF-10). Altitude is calculated from the measured pressure using the following international barometric formula:

$$altitude = 44330.0 \times \left(1.0 - \frac{P}{P_0}\right)^{0.190295}$$

  Where:
  P = pressure continuously measured by the BMP280 sensor
  P_0 = reference pressure obtained from the average of the first 5–10 pressure readings from the BMP280 at ground level.
- GPS: **GT-U7 module**. Dimensions: 16 × 12.2 × 2.4 mm. UART interface, 3.1–3.6 V. Supplies latitude, longitude, altitude, time, and satellite count (REF-11).
- Innovation – Wind sensor: **Anemometer Rev. C** (hot-wire). Dimensions: 17.2 × 40.386 × 6.36 mm. Operates at 5 V, 20–40 mA. Measures 0–60 mph using constant-temperature hot-wire method; highly sensitive to light gusts at 18–24 inches (REF-12).
- Microcontroller: **ESP32-S3-DevKitC-1** (dual-core 240 MHz, 44 GPIO, I2C/SPI/UART, 16 MB Flash).
- Telemetry: **RYLR998 LoRa** (primary) + **XBee 2.4 GHz** (secondary)
- Storage: **Micro-SD card module** (backup).
- Video recording: **RunCam Split 3 Micro** (proposed solution). HD 1080p video, autonomous operation (no programming required), 5 V supply, MOV format. Current draw 200–400 mA.

The entire electronic system, comprising sensors, radio frequency modules, storage modules, and a camera, is controlled by the ESP32 microcontroller. Therefore, the primary programming stack is the Espressif ESP-IDF program. Because it is programmed in C, it allows for the implementation of serial communication configurations (SPI, I2C, UART) with complete freedom to modify interfaces to meet user needs. It also allows the use of FreeRTOS to assign order, priority, and speed to tasks, provides a high level of modification and configuration of low-level configuration parameters in the modules, and enables the creation of custom component libraries to optimize resources, initialize them, and operate them according to the requirements of the call for proposals and the IR2030 regulations. Furthermore, it eliminates dependence on other pre-established libraries (or those with limited parameters) that may change over time or restrict configuration customization to some extent.

On the ground side, our team implemented a full-stack telemetry infrastructure. The local ground station (desktop application) was built with Electron and Node.js as the backend. We use the serialport library to communicate with the ESP32 receiver through the serial port, chart.js to generate real-time graphs of all sensor data, and IPC (Inter-Process Communication) to manage smooth data flow between the different tabs of the interface (dashboard, map, and 3D model).

The web dashboard (remote monitoring platform) uses the same chart.js library for consistent graphing, Three.js together with GLTFLoader to render a realistic 3D model of the CanSat (loaded from a .glb file), and is fully deployed on Cloudflare Pages. The backend API is hosted on Cloudflare Workers, where we use the native fetch API to handle GET and POST requests between the local station and the cloud. Finally, all telemetry is persistently stored in a SQLite database through Cloudflare D1, ensuring that data remains available even if the local application is closed.

The following is the detailed pre-launch action plan that our team will execute on the day of the official launch at Machrihanish. This sequence ensures that the payload is fully functional, calibrated, and ready before it is inserted into the launch vehicle. All steps are performed in the preparation area with at least two team members present for cross-checking.

Payload visual and mechanical inspection (T-3 h)

- Inspect the CanSat for physical damage, loose screws, or misaligned components.
- Verify that the anemometer ports are clear and the RunCam lens is clean.
- Confirm that the external power switch is in the OFF position and the SD card is properly seated.

Acceptance: No visible damage; all connectors secure.

Power system verification (T-2.5 h)

- Measure battery voltage with a multimeter.
- Connect the payload to the external power switch and confirm that all subsystems power up correctly (LED indicators on module are a good indicator).
- Run a 5-minute idle test to confirm total current draw is within the power budget.

Acceptance: Voltage ≥ 5.0 V, 3.3 V in all critical components and current draw ≤ 350 mA.

Firmware upload and functional test (T-2 h)

- Upload the latest firmware version via USB.
- Verify that the ESP32 boots correctly and all sensors initialise (GY-87, GT-U7, RevC, LoRa, XBee, SD card and RunCam).

- Perform a 10-minute continuous data-logging test: confirm that telemetry packets are generated every 3 seconds (or new cycle duty timelapse) and saved to SD card.

Acceptance: No boot errors and all sensors report valid data.

Sensor calibration and ground reference (T-1.5 h)

- Place the CanSat at the launch site and record, at least, 10 ground-level readings for BMP280 for setting P_0 reference and GPS lock.
- Calibrate the RevC against a handheld wind meter (must read within ±0.5 mph).
- Confirm GPS achieves ≤ 4 m horizontal accuracy.

Acceptance: P_0 stable (accuracy on altitude over ground), GPS locked, anemometer working.

Radio range and redundancy test (T-1 h)

- Perform a 150 m line-of-sight range test using both LoRa (869.5 MHz) and XBee (2.4 GHz) links.
- Verify packet reception rate ≥ 95 % at maximum distance.
- Confirm that the ground station (local Electron app + web dashboard) receives and displays all data correctly.

Acceptance: ≥ 95 % packet success on both links.

Video recording and synchronisation test (T-45 min)

- Start a 10-minute test recording with the RunCam Split 3 Micro.
- Verify that the video file is saved correctly to SD card and that telemetry timestamps match the video.

Acceptance: Video records at 1080p without interruption with timestamps align.

Final arming/disarming and integration test (T-15 min)

- Insert the CanSat into the launch-vehicle payload bay.
- Perform a full arm/disarm cycle using the external switch.

Acceptance: Payload remains powered and transmitted when armed.

This pre-launch sequence has been rehearsed during our ground tests and takes approximately 3 hours, leaving sufficient margin before the scheduled launch window for any inconvenience to need be solved.

The payload will be considered successful if the following metrics are satisfied:

- ≥ 95 % of telemetry packets successfully received via LoRa or XBee up to 450 m.
- Altitude measurement error ≤ 5 m compared with the launch-vehicle altimeter (verified by barometric formula).
- GPS fix achieved with horizontal accuracy ≤ 4 m.
- Minimum 5 minutes of continuous 1080p video recorded to SD card in MOV format (RunCam Split 3 Micro).
- All data (telemetry + video files) successfully recovered and viewable within 30 minutes after landing.

Finally, data will be recovered through three independent methods:

- First method: LoRa 869.5 MHz transmission to the ground station (local Electron app + web dashboard).
- Second method: XBee 2.4 GHz transmission to the ground station (local Electron app + web dashboard).
- Third method: Cloud backup visible on the website.
- Fourth method: Recovery data from the micro-SD.

All telemetry samples are automatically forwarded to a Cloudflare D1 database via the ground station Worker API, creating a persistent cloud backup. Video files will be copied from the SD card and uploaded to the same cloud repository as a backup of the file. The four-method approach should ensure 100 % data availability even in the event of a single-link failure.

3.5 Requirements

Functional Requirements

ID	Description	Justification	Verification Method
FUN-1	Acquire altitude, GPS position, IMU data, wind speed and video continuously during flight.	Mandatory sensors and video recording required by Mach-26 Category 1 (REF-13).	Test and Demonstration
FUN-2	Transmit telemetry in real time using LoRa primary + XBee secondary as a links	Ensures reliable data delivery and fulfils the mandatory backup tracking system.	Test
FUN-3	Store all telemetry and video data locally on Micro-SD card.	Provides tertiary data recovery in case of radio link failure.	Test
FUN-4	Calculate and transmit altitude using barometric formula in real time.	Required for accurate apogee detection and drift analysis.	Analysis and test

Performance Requirements

ID	Description	Justification	Verification Method
PER-1	Achieve ≥ 95 % successful packet reception up to 450 m altitude.	Guarantees reliable real-time telemetry for scoring and safety monitoring.	Analysis and test
PER-2	Maintain GPS horizontal accuracy ≤ 4 m.	Required for precise drift calculation and landing position.	Test

PER-3	Record minimum 5 minutes of continuous 1080p video to SD card.	Satisfies the Category 1 video recording challenge.	Test
PER-4	Transmit wind-speed data (0–60 mph) with update rate of at least 2 seconds.	Innovation requirement and scientific value of the anemometer.	Test
PER-5	The quail egg must remain intact after landing	Demonstrates the cushioning system's effectiveness in transporting fragile cargo	Test and demonstration

Design Requirements

ID	Description	Justification	Verification Method
DES-1	The assembly of the payload can be put together without applying excessive force to its components	It demonstrates sound structural design and appropriate component selection	Demonstration
DES-2	The CanSat weight is ≤ 350 gr and its dimentions don't surpass the stated in Figure 7	In accordance with these measures, it occupies no more than the space allocated to it within the rocket and, as such, does not affect the lauch vehicle's flight stability or assembly	Test
DES-3	The parachute must demonstrate fall rates consistent with those shown in the descent control section	Verifying that the design and calculations were accurate, ensuring the structural integrity of the CanSat	Test and analysis

Interface Requirements

ID	Description	Justification	Verification Method
INT-1	All sensors communicate with ESP32-S3 via standard protocols (I2C, UART, SPI).	Guarantees reliable and modular integration of GY-87, GT-U7, anemometer and LoRa.	Test
INT-2	External power switch allows safe arming/disarming without opening CanSat.	Required for safe operations on the launch pad.	Demostration
INT-3	Telemetry data format compatible with both local	Enables seamless real-time display and cloud backup.	Test

	Electron station and web dashboard.		

Operational Requirements

ID	Description	Justification	Verification Method
OP-1	Payload must be fully testable and armable within 3 hours prior to launch.	Matches the pre-launch testing plan (Section 3.4).	Demostration
OP-2	SD card and video card must be easily removable after landing.	Allows rapid data recovery by the recovery team.	Demostration

Additional Requirements

ID	Description	Justification	Verification Method
ADD-1	Provide cloud backup of all telemetry via Cloudflare D1.	Adds robustness and allows remote monitoring by judges and team.	Test
ADD-2	Operate reliably in Machrihanish wind and rain conditions.	Ensures functionality under expected launch-site weather.	Test

4. Testing

4.1 Airframe Tests

The objective of the airframe testing campaign is to validate the structural integrity of the launch vehicle under expected aerodynamic and mechanical loads experienced during ascent and recovery.

4.1.1 Structural Load Testing

A static load test will be conducted to validate the structural resistance of the airframe and fin assembly under equivalent aerodynamic loading conditions.

Method:

- Apply distributed loads on the fin surfaces equivalent to calculated Max-Q conditions (~6500 Pa from Section 2.3)
- Apply bending loads at the fin root to simulate aerodynamic moments
- Measure deformation and check for structural failure

Objective:

- Verify compliance with **DES-1 and DES-3 requirements**
- Ensure no permanent deformation or cracking occurs

4.1.2 Fin Strength and Flexural Testing

To validate the laminated cedar fin structure:

Method:

- Perform a **3-point bending test** on representative fin samples
- Measure maximum load before failure
- Compare results with analytical predictions

Objective:

- Validate resistance to aeroelastic loads
- Confirm adequate stiffness to prevent flutter

4.1.3 Assembly Integrity Test

A full airframe inspection and fit-check will be conducted:

Method:

- Assemble full vehicle
- Verify alignment of fins, motor mount, and payload bay
- Check structural bonding and adhesive joints

Objective:

- Validate manufacturing quality
- Ensure compliance with interface requirements

4.2 Recovery Tests

The recovery system testing aims to validate reliable parachute deployment and safe descent behaviour under realistic conditions.

4.2.1 Parachute Deployment Test

Method:

- Ground-based ejection simulation using compressed air or test charge
- Validate nosecone separation and parachute extraction

Objective:

- Verify **FUN-1 and INT-1 requirements**
- Ensure no entanglement or deployment failure

4.2.2 Descent Rate Validation

Method:

- Drop test from elevated structure (≥ 10 m recommended)

- Measure descent velocity using video analysis

Objective:

- Validate analytical descent velocity (~10–12 m/s)
- Confirm compliance with **PER-3 requirement**

4.2.3 Shock Cord Load Test

Method:

- Apply tensile load equivalent to deployment shock
- Measure elongation and inspect for failure

Objective:

- Validate structural integrity of Kevlar cord
- Ensure resistance to early deployment loads (~12.1 m/s)

4.3 Payload Deployment Tests

Method:

- Low altitude (< 100 m) test launch using low power launch vehicule with comparable dimensions
- Observe separation between airframe and CanSat
- Analyse payload descent and ejection

Objective:

- Validate analytical payload descent velocity
- Confirm payload correct ejection
- Test effectiveness of the egg's cushioning system during the mock flight

4.4 Payload Ejection Tests

This testing phase focuses on validating the safe and reliable separation of the CanSat payload during the recovery event.

4.4.1 Ejection Separation Test

Method:

- Simulate ejection event using ground-based pressurisation
- Observe separation between airframe and CanSat.

Objective:

- Verify clean separation without collision or re-contact
- Validate compliance with **FUN-1 requirement**

4.4.2 Deployment Dynamics Test

Method:

- Record high-speed video of deployment
- Analyse trajectory and separation distance

Objective:

- Ensure both systems deploy independently
- Prevent parachute entanglement

4.4.3 Integrated Recovery Test

Method:

- Full system test including:
  - Ejection
  - Separation
  - Dual parachute deployment

Objective:

- Validate complete recovery sequence
- Identify system-level interactions

4.4 Payload Electronics Tests

The payload electronics testing aims to validate the operational autonomy and mechanical robustness of the onboard electronic system under realistic mission and extreme vibration conditions.

4.5.1 Power Endurance and Communication Test

Method:

- Connect all payload electrical loads in their operational configuration
- Power the complete system using the flight battery
- Maintain continuous communication and data transmission for as long as possible
- Monitor battery voltage, current consumption, and system response throughout the test
- Record the time at which the battery voltage drops below the minimum level required for stable operation

Objective:

- Validate the effective autonomy of the payload electronics under realistic load conditions
- Verify that the communication link remains stable during extended operation
- Identify the practical operating time before undervoltage affects performance
- Confirm that the electronics can sustain mission-critical functions for the required duration

4.5.2 Vibration Endurance Test

Method:

- Install the CanSat with the PCBs mounted inside the structure in its representative flight configuration
- Place the system on the vibration platform
- Subject the payload electronics to prolonged vibration exposure at different frequencies
- Continuously monitor system behaviour during vibration, including communication status, resets, voltage fluctuations, sensor response, and possible intermittent failures
- Inspect the electronics and internal connections after the test for any signs of loosening, damage, or malfunction

Objective:

- Evaluate the robustness of the payload electronics under severe mechanical vibration
- Identify failures caused by long-duration excitation at different frequencies
- Verify the integrity of PCB mounting, connectors, solder joints, and wiring under dynamic loading
- Confirm that the system can continue operating and communicating while exposed to intense vibration environments

4.5 Payload Recovery Tests

Method:

- Drop test from a drone at various altitudes (50 m – 100 m max. due to urban environment)
- Measure descent velocity using video analysis

Objective:

- Validate analytical descent velocity (< 4 m/s)
- Confirm compliance with payload **DES-3** requirement
- Test effectiveness of the egg's cushioning system

4.7 Any other tests specific to your CanSat and/or Launch Vehicle.

4.7.1 Ground Station Local Software Integration Tests

This testing phase focuses on validating the reception, parsing, visualisation and local persistence behaviour of the desktop ground station when connected to the ESP receiver.

Method:

- Connect the ESP-based receiver to the laptop through the operational USB serial interface

- Transmit representative telemetry packets through LoRa and XBee so that the ESP forwards them through the same serial channel
- Verify that the Electron ground station identifies the selected port, parses incoming messages and updates the dashboard, map and 3D view correctly
- Generate a local report and confirm that the received telemetry is included in the exported files

Objective:

- Validate that the local ground station software can operate as the primary mission interface without manual intervention
- Confirm that telemetry reception from the ESP concentrator is compatible with the parser implemented in the station

Status: In progress. Parser compatibility tests have already been implemented in software and real serial integration tests are planned with the flight hardware.

4.7.2 Telemetry Parser and Serial Interface Tests

Method:

- Execute automated parser tests using tagged messages such as [LORA] LAT:...,LON:... and [XBEE] LAT:...,LON:... with a partial system
- Test mixed messages containing TX, RX and distance information to validate transmitter-receiver separation calculations
- Inject invalid or noisy log lines to confirm that the station ignores non-telemetry serial output

Objective:

- Validate that the ground station can safely process partial telemetry during the current project phase
- Ensure future compatibility with complete tagged telemetry once all payload magnitudes are transmitted

Status: Completed at software level. Automated parser tests currently pass for LoRa, XBee.

4.7.3 Web Ground Station and Cloud API Tests

Method:

- Send telemetry from the local ground station to the Cloudflare Worker using the operational HTTP endpoint
- Query /api/health, /api/latest, /api/recent and /api/report to verify endpoint behaviour and response integrity
- Open the web ground station from multiple clients and verify that dashboard plots, map trajectory, 3D model and report download update correctly

Objective:

- Validate that the remote ground station reflects the same mission state as the local Electron station
- Verify that the Worker API remains compatible with real-time visualisation and report export features
- Status: Partially completed. Deployment, endpoint validation and web visualisation have already been verified. Multi-user operational tests remain pending.

4.7.4 Database backup and persistence Tests

Method:

- Insert telemetry into the Worker API and confirm storage in the Cloudflare D1 database
- Validate that the latest sample, recent history and downloadable reports remain available after multiple requests and session changes
- Confirm that additional fields such as source channel, receiver coordinates and transmitter-receiver distance persisted correctly

Objective:

- Verify that the cloud backup system can preserve telemetry independently of the local desktop application state
- Ensure that the remote web station always reads from a persistent and consistent data source
- Status: Partially completed. D1 migrations and endpoint persistence were validated, but longer-duration mission tests remain to be completed.

4.7.5 End-to-End Redundancy and Communications Tests

Method:

- Use a both LoRa and XBee links through the ESP receiver for field testing in various situations
- Compare whether the station continues to receive valid position information when one channel degrades or becomes intermittent
- Measure whether the calculated distance, trajectory and speed remain coherent during handover or mixed reception conditions

Objective:

- Validate the redundancy philosophy of the communications subsystem at the ground segment level
- Demonstrate that the mission can continue receiving useful telemetry even if one wireless link becomes unreliable

Status: Partially realized. Initial tests have been carried out, but more intensive testing is planned.

5. Safety Management

5.1 Safety Compliance

Our mission has been designed using a systems engineering approach that prioritises physical and operational integrity, in strict compliance with the UKRA Safety Code and the technical requirements of the Mach-26 organising committee.

1) Mach-26 Guidelines Requirement Compliance
    - Propulsion and Category: The team is competing in Category 1 using a commercial off-the-shelf (COTS) Aerotech G80T-7 solid rocket motor. Its total thrust remains below the maximum limit of 160 Ns set for this class.
    - Structural Integrity and Retainer (ASA): The main fuselage is made from high-strength phenolic cardboard (kraft paper). The motor is secured by a 3D-printed retainer using ASA (Acrylonitrile Styrene Acrylate) filament. This material is ideal for aerospace use due to its high heat deflection temperature (HDT), which ensures its integrity against the heat radiated by the motor. Mechanically, its high toughness enables it to meet the requirement of withstanding and transmitting a force equivalent to twice the engine's maximum thrust without suffering permanent deformation, whilst also outperforming other polymers in terms of durability against UV radiation and humidity in the flight environment.

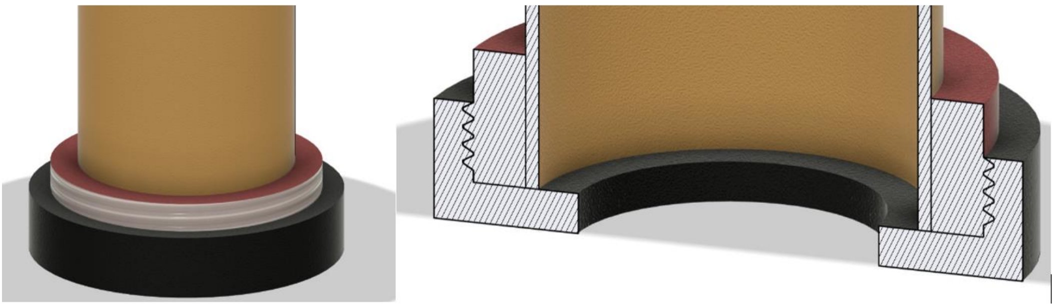


Fig. 14. 3D printing retainer for ASA filament to hold the motor

    - Ignition and Ramp Safety Protocol: To avoid unsafe redundancies with the organisation's systems, the rocket does not incorporate a proprietary pull-pin disarming cable. Instead, the system relies entirely on the event's launch protocols, utilising a three-point safety mechanism operated exclusively by the Range Safety Officer (RSO).
    - Recovery System: Cone separation and payload ejection are performed mechanically via the engine's secondary ejection charge. To prevent burns to the parachute and structure caused by combustion gases, a fire-retardant covering is used.
    - Descent Performance and Altitude Measurement: A vertical landing speed of 12 m/s has been validated through calculations, which meets the requirement to maintain the speed below 15 m/s. To comply with the

mandatory apogee recording (~450 m), our design utilises the CanSat's avionics. The GY-87 module is used, which incorporates a high-precision BMP280 barometer. The telemetry data collected during the ascent, up to mechanical ejection, will provide the official mission reading.

- Redundant Tracking and Video Challenge: The CanSat features a highly robust dual-path telemetry architecture, operated by an ESP32-S3 microcontroller. The primary link uses LoRa radio frequency (RYLR998 at 869.5 MHz), whilst backup tracking is carried out via the XBee 3 protocol (2.4 GHz). Positioning is obtained via a GT-U7 GPS module. Additionally, to meet the category challenge, the CanSat records video in HD format using a RunCam Split 3 Micro, storing the data locally on an SD card without saturating the radio frequency bandwidth.

Note on Independent Tracking and Engagement with the Committee: As our separation sequence is purely mechanical and we have omitted a flight computer from the fuselage to optimise mass, the rocket body initially relies on visual tracking of its parachute. The CanSat has its own GPS and telemetry system for autonomous tracking. Our team acknowledges the rule requiring a backup tracking system for the vehicle; therefore, if following the technical review the committee determines that the inclusion of an electronic system in the rocket is strictly necessary and mandatory, we undertake to adapt and integrate an autonomous tracking module (RF/GPS) prior to the Flight Readiness Review (FRR).

2) Team Requirement Compliance

The team operates within the established membership limit and is supervised by an academic liaison at our university. Each member has clearly defined responsibilities and must verify the integrity of their respective subsystem prior to the final safety inspection at the event.

5.2 Risks and Mitigation

The risk assessment has been thoroughly updated since the PDR phase, identifying potential issues and setting out clear responsibilities to ensure the mission's success.

ID	Risk	Impact	Probability	Severity	Mitigation
1	Ignition failure or engine delay.	Unstable launch or mission abort.	Low	High	Exclusive use of a certified COTS engine. Continuity checks during the ramp-up phase and strict adherence to safety protocols and authorisation from the Mach-26 RSOs.
2	Structural failure of the fuselage, fins or retainer.	Loss of the vehicle and the generation of hazardous waste.	Low	High	Made from reinforced cardboard tubing (Apogee Rockets), cedar wood fins/rings and an ASA-printed

					retainer. Pre-launch inspections to check for cracks or severe deformation, and static load tests to ensure the structure can withstand twice the maximum thrust.
3	Malfunction of the recovery system (failure to deploy).	Hard landing (>15 m/s), damage to the payload and the rocket.	Low	High	Ground tests of the deployment mechanism with the engine's ejection load. Reinforcement of connections with Kevlar cord to absorb impacts, and use of fire-retardant fabric.
4	Failure to separate the CanSat.	Loss of payload data and an unsafe descent.	Low	Medium	A simple mechanical separation design driven by engine gases. Multiple fit checks and ground deployment tests.
5	Failure of the telemetry or tracking system.	Dificultad para localizar el CanSat tras el aterrizaje.	Medium	Low	Redundant telemetry architecture (LoRa + XBee) and on-board data logging via SD card. (Installation of a backup RF module on the rocket if required).
6	Adverse weather conditions (wind, rain).	Drift outside the permitted recovery zone or electrical damage.	Medium	Medium	Pre-flight weather monitoring, waterproof encapsulation of the avionics, and adherence to the Mach 26 holdback criteria against adverse weather

Failure Modes Matrix

Subsystem	Failure mode	Effect on the mission	Prevention/verification method
Fuselage	Deformation of the kraft paper tube caused by moisture or thermal stress.	Inestabilidad aerodinámica y posible desviación de la trayectoria.	Surface treatment with phenolic sealing; structural rigidity tests.
Recovery	Breakage or tangling of the Kevlar cord.	Separation of the cone and an unstable descent of the rocket at high speed.	Design for extreme shock loads; use of swivels and packaging protocol.
CanSat Electronics	Fault in pressure sensors or loss of telemetry signal.	Inability to record apogee altitude and other data.	Vibration testing on the ground and reinforcement of joints/welds using thermal adhesive.
Payload	Mechanical jam of the CanSat inside the rocket fuselage.	Failure of the experiment and parachutes to deploy.	Polishing the inside of the tube and carrying out multiple mechanical fit checks.
Deploy	Premature separation of the cone during take-off.	Premature activation of the critical-speed recovery system.	Installation of shear pins designed to withstand in-flight internal pressure.

6.Project Management

6.1 Schedule

In the following image, the Work Breakdown Structure (in a phase-based format) of the project is presented.

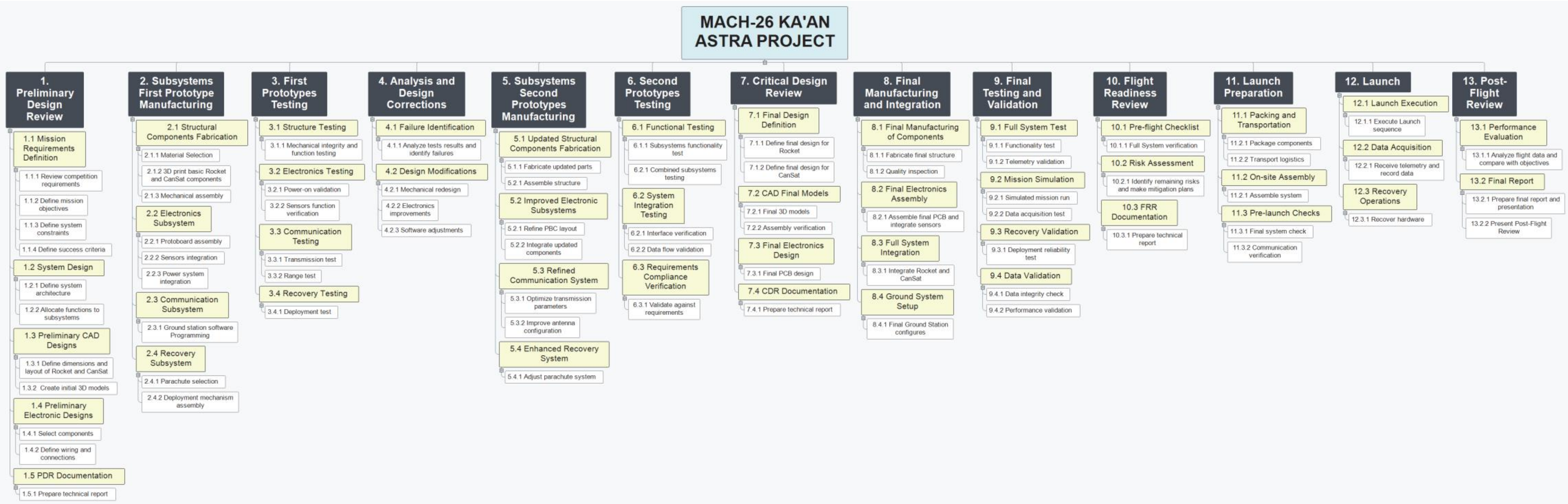


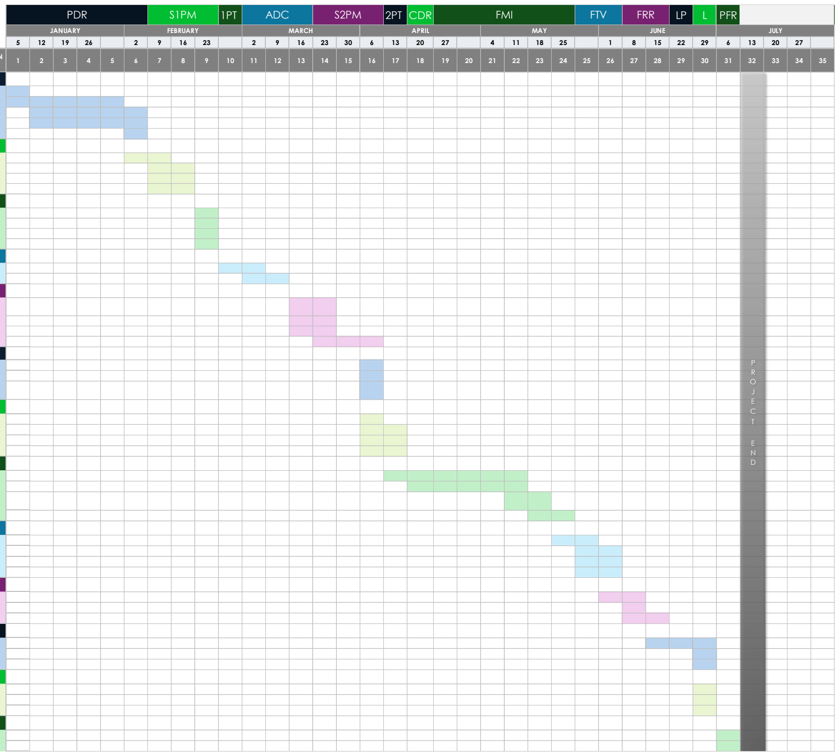
Fig 15. Work Breakdown Structure of the Project.

Gantt Chart

Based on the WBS, a Gantt chart has been developed including the key tasks and the Level 2 subtasks.

MACH-26 KA'AN ASTRA PROJECT

TASK	DEPENDENCY	START DATE	END DATE	DURATION (DAYS)
1. Preliminary Design Review (PDR)				
1.1 Mission Requirements Definition		02/01/2026	06/01/2026	4
1.2 System Design	1.1	07/01/2026	12/01/2026	5
1.3 Preliminary CAD Designs	1.1, 1.2	12/01/2026	01/02/2026	20
1.4 Preliminary Electronic Designs	1.1, 1.2	13/01/2026	02/02/2026	20
1.5 PDR Documentation	1.1, 1.2, 1.3, 1.4	03/02/2026	06/02/2026	3
2. Subsystems First Prototype Manufacturing (S1PM)				
2.1 Structural Components Fabrication	1.3	07/02/2026	15/02/2026	8
2.2 Electronics Subsystem	1.4	08/02/2026	23/02/2026	15
2.3 Communication Subsystem	1.2	09/02/2026	23/02/2026	14
2.4 Recovery Subsystem	PDR	10/02/2026	23/02/2026	13
3. First Prototypes Testing (1PT)				
3.1 Structure Testing	2.1	24/02/2026	25/02/2026	1
3.2 Electronics Testing	2.2	24/02/2026	25/02/2026	1
3.3 Communication Testing	2.3	25/02/2026	26/02/2026	1
3.4 Recovery Testing	2.4	25/02/2026	26/02/2026	1
4. Analysis and Design Corrections (ADC)				
4.1 Failure Identification	1PT	27/02/2026	04/03/2026	5
4.2 Design Modifications	4.1	04/03/2026	16/03/2026	12
5. Subsystems Second Prototypes Manufacturing (S2PM)				
5.1 Updated Structural Components Fabrication	4.2	17/03/2026	28/03/2026	11
5.2 Improved Electronic Subsystems	4.2	17/03/2026	28/03/2026	11
5.3 Refined Communication System	4.2	17/03/2026	28/03/2026	11
5.4 Enhanced Recovery System	4.2	29/03/2026	07/04/2026	9
6. Second Prototypes Testing (2PT)				
6.1 Functional Testing	S2PM	08/04/2026	09/04/2026	1
6.2 System Integration Testing	S2PM	08/04/2026	09/04/2026	1
6.3 Requirements Compliance Verification	S2PM	09/04/2026	10/04/2026	1
7. Critical Design Review (CDR)				
7.1 Final Design Definition	2PT	11/04/2026	13/04/2026	2
7.2 CAD Final Models	7.1	13/04/2026	15/04/2026	2
7.3 Final Electronics Design	7.1	13/04/2026	15/04/2026	2
7.4 CDR Documentation	7.1, 7.2, 7.3	15/04/2026	17/04/2026	2
8. Final Manufacturing and Integration (FMI)				
8.1 Final Manufacturing of Components	CDR	18/04/2026	14/05/2026	26
8.2 Final Electronics Assembly	8.1	21/04/2026	06/05/2026	15
8.3 Full System Integration (Rocket + CanSat + Ground Station)	8.1, 8.2	15/05/2026	23/05/2026	8
8.4 Ground System Setup		21/05/2026	25/05/2026	4
9. Final Testing and Validation (FTV)				
9.1 Full System Test	FMI	26/05/2026	28/05/2026	2
9.2 Mission Simulation	9.1	30/05/2026	05/06/2026	6
9.3 Recovery Validation	9.1	30/05/2026	05/06/2026	6
9.4 Data Validation	9.4	30/05/2026	05/06/2026	6
10. Flight Readiness Review (FRR)				
10.1 Pre-flight Checklist	FTV	06/06/2026	10/06/2026	4
10.2 Risk Assessment	10.1	11/06/2026	14/06/2026	3
10.3 FRR Documentation	10.1, 10.2	12/06/2026	19/06/2026	7
11. Launch Preparation (LP)				
11.1 Packing and Transportation	FRR	20/06/2026	28/06/2026	8
11.2 On-site Assembly	11.1	29/06/2026	30/06/2026	1
11.3 Pre-launch Checks	11.1, 11.2	29/06/2026	30/06/2026	1
12. Launch (L)				
12.1 Launch Execution	LP	29/06/2026	01/07/2026	2
12.2 Data Acquisition	12.1	29/06/2026	01/07/2026	2
12.3 Recovery Operations	12.1	29/06/2026	01/07/2026	2
13. Post-Flight Review (PFR)				
13.1 Performance Evaluation	L	01/07/2026	03/07/2026	2
13.2 Final Report	13.1	01/07/2026	03/07/2026	2



6.2 Budget

The following tables show the updated itemized expenses of the mission; the ***estimated budget*** is around ***£13 650***.

MANUFACTURING				
Material	**Unit**	**Unit Value ($)**	**Unit Value (MXN$)**	**Subtotal (MXN$)**
CanSat				
Filament (PETG with carbon fiber)	2	-	$ 919.00	$ 1,838.00
Phenolic board (for PCB)	6	-	$ 19.00	$ 114.00
GPS	1	-	$ 422.80	$ 422.80
GY-87 Sensor	2	-	$ 221.00	$ 442.00
Xbee Module XB3-24AUT-J	3	$ 30.58	$ 530.87	$ 1,592.61
LoRa Module RYLR998	3	-	$ 459.22	$ 1,377.66
Voltage Regulator (5 V)	2	-	$ 24.00	$ 48.00
Voltage Regulator (3,3 V)	2	-	$ 28.00	$ 56.00
ESP32-s3	2	-	$ 129.78	$ 259.56
Solder Wire (Tin)	1	-	$ 50.00	$ 50.00
Wire	1	-	$ 5.00	$ 5.00
Camera	1	-	$ 2,633.00	$ 2,633.00
Antenna ANTX100P001B24003	3	$ 3.55	$ 21.30	$ 63.90
Antenna SRF2W012-150	3	$ 5.68	$ 5.68	$ 17.04
			SUBTOTAL (MXN$)	$ 8,919.57
ROCKET				
Body Tube		$ 28.34		$ 491.98
Tube Coupler		$ 5.49		$ 95.31
Shock Cord		$ 18.80		$ 326.37
Rail Adapters		$ 26.58		$ 461.43
Parachute		$ 7.86		$ 136.45
Motor		$ 116.97		$ 2,030.60
			SUBTOTAL (MXN$)	$ 3,542.13
			TOTAL (MXN$)	**$ 12,461.70**

TRIP				
Expenses	**Unit**	**Unit Value (€)**	**Unit Value (MXN$)**	**Subtotal (MXN$)**
Entrance (General)	15	£ 13.33	$ 316.85	$ 4,752.81
Entrance (Extra)	11	£ 100.00	$ 2,377.00	$ 26,147.00
Flight (Cancun-Londres)	10	-	$ 23,000.00	$ 230,000.00
Train (Londres-Glasgow)	11	-	$ 2,500.00	$ 27,500.00
Accommodation (Beds)	10	£ 80.00	$ 1,901.60	$ 19,016.00
Accommodation (Hostel)	1	£ 65.00	$ 1,545.05	$ 1,545.05
Senior BMFA Membership	2	£ 51.00	$ 1,212.27	$ 2,424.54
			TOTAL (MXN$)	$ 311,385.40

TOTAL (MXN$)	$ 323,847.10
TOTAL (£)	$ 13,624.19

Sponsorship

The mission is currently supported by sponsorships from ***Universidad Autónoma de Yucatán (UADY)***, local electronics suppliers ***Thido*** and ***Steren*** (for some components), **All Blue** (a games store), **and CEAE** (an academic advising centre). In addition to these partnerships, the team has launched a **GoFundMe** campaign to obtain community-based donations to help us to complete this mission. The team continues its efforts to secure sponsorship by expanding collaborations with educational organizations through direct outreach, promotional activities, and mission-related visibility opportunities.

7. Conclusion

This Critical Design Review demonstrates that the Mach-26 project has progressed from a preliminary concept into a substantially integrated and technically justified system. Since the PDR, the team has refined the mission definition, consolidated the launch vehicle and CanSat architectures, and strengthened the link between design decisions, analytical validation, and operational safety. The current design shows that the project is aligned with the main mission objectives: achieving a target apogee close to 450 m, ensuring safe recovery of both the launch vehicle and the CanSat, acquiring and storing flight data, recording onboard video, and safely transporting the additional fragile payload.

The major accomplishments since the PDR are substantial. On the launch vehicle side, the team completed an important structural and mass optimisation process, achieving a 51 g mass reduction and a final launch-ready mass of 1091 g. The fin system was redesigned into a quasi-isotropic cedar laminate to improve stiffness and flutter resistance, while the nosecone was redefined as a controlled 5 mm PETG shell to improve strength and mass balance. These updates resulted in a stable configuration with a static margin of 1.84 calibres, an off-rail velocity of 13.1 m/s, and a predicted apogee of 476 m, all supported by flight analysis. In parallel, the propulsion procurement risk was reduced by confirming the primary motor order with a UKRA-approved vendor and defining a compatible backup motor strategy. On the payload side, the CanSat evolved into a more robust and environmentally protected design: the enclosure was sealed, the PCB count was reduced from three to two, the battery was relocated for better mass distribution, and a dedicated camera compartment was integrated. The quail egg protection system was also defined through a cushioning concept based on polystyrene beads and TPU confinement. In addition, the team developed a more complete electrical architecture and a redundant telemetry chain based on LoRa, XBee, onboard SD storage, a local Electron ground station, and a Cloudflare-based remote dashboard. Together, these developments represent a clear increase in design maturity, subsystem compatibility, and mission robustness compared to the PDR phase.

The main unfinished work is no longer related to the general concept of the mission, but rather to experimental validation, final integration, and operational closure. Several physical tests remain to be completed, including structural tests, parachute deployment and descent validation, payload ejection tests, payload electronics endurance testing, vibration testing, and full mission simulation. On the ground segment, although the parser logic and parts of the web telemetry infrastructure have already been validated, complete real-hardware serial integration, multi-user web verification, longer-duration persistence tests, and more demanding field redundancy tests are still pending or only partially completed. In addition, the final budget section still requires final polishing, and there remains a conditional action regarding the possible addition of an autonomous tracking module for the rocket body if requested by the technical review committee before the FRR.

To resolve these remaining tasks, the team has already defined a structured plan in the project schedule. The next steps include payload electronics and communications validation, local and remote end-to-end testing, distance and stress testing of both communication links, full mission simulation including recovery and telemetry, and a final review focused on closing test evidence and implementing corrections. This staged approach reduces integration risk and allows the conversion of the current analytical and design maturity into verified flight readiness.

8. Acknowledgements

The team acknowledges the Universidad Autónoma de Yucatán for the economic support provided, the enterprise Mantenimiento y Manufacturas Industriales Zambrano S.A. de C.V. for giving us acces to their equipment, to the kind persons who donated to our cause in Go Fund Me and the shop All Blue for promoting the team.

References

ID	Document Title	Date of Issue	Link
REF-1	IR 2030 – UK Interface Requirements 2030	2023	https://www.ofcom.org.uk/siteassets/resources/documents/spectrum/interface-requirements/ir-2030.pdf?v=415472

REF-2	RYLR998	2026	https://reyax.com/upload/products_download/download_file/RYLR998_EN.pdf
REF-3	LoRa® and LoRaWAN®	2024	https://www.semtech.com/uploads/technology/LoRa/lora-and-lorawan.pdf
REF-4	LoRa AT COMMAND GUIDE	2021	https://reyax.com/upload/products_download/download_file/LoRa_AT_Command_RYLR998_RYLR498_EN.pdf
REF-5	SX1276/77/78/79 - 137 MHz to 1020 MHz Low Power Long Range Transceiver	2015	https://cdn-shop.adafruit.com/product-files/3179/sx1276_77_78_79.pdf
REF-6	DIGI XBEE 3 802.15.4	2021	https://digi.com/resources/library/data-sheets/ds_xbee3-802-15-4
REF-7	Digi XBee® 3 802.15.4. Radio Frequency (RF) Module	2023	https://docs.digi.com/resources/documentation/digidocs/pdfs/90002273.pdf
REF-8	EU Declaration of Conformity	2025	https://ftp1.digi.com/support/documentation/certifications/DoC,%20XBee3,%20RED,%20ML,%20Cyber,%200925.pdf
REF-9	UK Declaration of Conformity	2022	https://ftp1.digi.com/support/documentation/certifications/UK_DoC,%20XBee3,%20UKCA,%201222.pdf
REF-10	GY-87 10DOF MPU6050 HMC5883L BMP180 Sensor Module	n.d.	https://uelectronics.com/wp-content/uploads/2018/10/gy-87-10dof-mpu6050-hmc5883l-bmp180-sensor-module.pdf
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REF-12	Wind Sensor Rev. C	2026	https://moderndevice.com/products/wind-

			sensor?srsltid=AfmBOopFcK6h3bXXW6CDrogjDawQTBkBtSaPb2KbsAxxtEMnfcxbrjqN
REF-13	MACH-26 LAUNCH COMPETITION & CONFERENCE. RULES & REQUIREMENTS.	2026	https://0fb2ba8d-01fc-4cae-8fc3-e304b69d2b72.filesusr.com/ugd/679ee3_e038af0bdb12414781cf27a6bd48a762.pdf
REF-14	Report of the E.L.V.I.R.A Mark I	2025	
REF-15	Parachute cost, types, weight limits: Everything you need to know	2024	https://skydivepalmbeach.com/blog/parachute-cost-types-weight-limits/
REF-16	Computational fluid dynamic (CFD) analysis of parachute canopies design for Aludra SR-10 UAV as a Parachute Recovery Systems (PRS)	2020	https://www.akademiabaru.com/doc/CFDLV12_N2_P46_57.pdf
REF-17	Study applied to the descent of a parachute-load system and calculation of its trajectory	2018	https://hdl.handle.net/10016/29307
REF-18	Analysis of the stability of a parabolic parachute with chimney retraction	2016	https://www.researchgate.net/publication/363862539_Stability_analysis_of_a_vent_reefed_parabolic_parachute_in_Spanish
REF-19	A two-dimensional mathematical model of a parachute in steady descent (C.P. No. 1260)	1973	https://reports.aerade.cranfield.ac.uk/handle/1826.2/1273